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Geochemistry and genesis of magmatic Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China

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ABSTRACT

Magmatic sulfide deposits can be subdivided into 1) Ni-Cu-PGE deposits, which have relatively smooth mantle-normalized metal patterns, 2) Ni-Cu-(PGE) deposits, which are depleted in PGE relative to Ni-Cu-Co, and 3) PGE-(Cu)-(Ni) deposits, which are enriched in PGE relative to Ni-Cu-Co. China contains almost exclusively deposits in the second group, including the world's largest semi-continuous ore body (Jinchuan) and the largest known deposit in a possible arc setting (Xiarihamu), but which are less common elsewhere (e.g., Pechenga, Voisey's Bay). Unlike other magmatic Ni-Cu-PGE, Ni-Cu-(PGE), and PGE-(Cu)-(Ni) deposits, many of which are Neoarchean or Paleo-Mesoproterozoic and formed primarily in rifted-related settings, all known Chinese deposits are younger (some Neoproterozoic, but mainly mid-late Paleozoic) and many are inferred to have formed in settings that previously experienced subduction. Based on mineralization age, tectonic setting, and spatial distribution, most deposits occur in 3 tectono-magmatic settings: 1) Neoproterozoic belts related to the breakup of the Rodinian supercontinent (e.g., Jinchuan, Zhouan); 2) Devonian to Triassic magmatism in the Central Asian (CAOB) and East Kunlun (EKOB) orogenic Belts (e.g., Huangshan, Hongqiling, Kalatongke, Xiarihamu), and 3) the late Permian Emeishan large igneous province (ELIP) (e.g., Jinbaoshan, Zhubu, Baimazhai). Many (Huangshandong, Huangshannan, Huangshanxi, Jinchuan, Jingbulake, Kalatongke #1, Hongqiling #1, Limahe, Qingkuangshan, Zhubu) are hosted by small intrusions with diamond-shaped surface sections and funnel-shaped cross sections that have been interpreted to represent subvertical transtensional structures, but which are asymmetrically differentiated and are more likely sections through subhorizontal blade-shaped dikes. A few are hosted by subhorizontal chonoliths (e.g., Kalatongke #2). Only a few are hosted by subhorizontal sills (e.g., Jinbaoshan, Yangliuping). Mineral chemical, whole-rock lithogeochemical, ore geochemical, and S-Nd-Sr-Os isotopic data for 18 typical deposits have been used to aid in the assessment of their genesis and metallogeny. Most deposits in orogenic belts appear to be hosted by rocks derived from magmas generated from subduction-enriched, but originally depleted mantle sources with minor crustal contamination. Most deposits in the ELIP appear to be hosted by rocks derived from magmas generated from subduction-enriched, but originally more enriched mantle sources with variable degrees of crustal contamination. Deposits related to the breakup of Rodinia exhibit transitional geochemical characteristics. Relatively high Ni-Cu-Co and relatively low PGE tenors, high-Ni in olivine at a given Fo content, high γOs, and intermediate εNd values suggest that many Chinese Ni-Cu-(PGE) deposits were derived by melting Ni-Co-Cu-rich PGE-poor pyroxenitic mantle, most likely produced by interaction of recycled oceanic crust with depleted mantle peridotite. Variable PGE tenors that correlate inversely with δ^{34}S and γOs values suggest that most deposits formed at low-moderate (10–1000) magma:sulfide mass ratios (R factors). Some deposits exhibit fractionations of Ni_{100}-Co_{100}-$IPGE_{100}$ from Cu_{100}-Au_{100}-$PPGE_{100}$ (metals in 100% sulfides) indicating that the sulfide melts experienced variable degrees of MSS fractionation/accumulation. Compared to Archean and Proterozoic magmatic Ni-Cu-PGE deposits elsewhere in the world, most of which appear to have formed primarily in rifted continental and rifted continental margin settings and to have been derived from peridotitic mantle, most of the Phanerozoic Ni-Cu-(PGE) deposits in China appear to have formed in convergent or formerly convergent settings and to have had variable amounts of metasomatized pyroxenitic mantle in their sources.

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1. Introduction

Magmatic sulfide deposits can be subdivided into 1) Ni-Cu-PGE deposits, which have relatively smooth mantle-normalized Ni-Cu-Co-PGE metal patterns (e.g., Duluth, Kambalda-Mt Keith-Perseverance, Noril'sk-Talnakh, Raglan, Sudbury, Thompson), 2) Ni-Cu-(PGE) deposits, which are depleted in PGE relative to Ni-Cu-Co (e.g., Jinchuan, Pechenga, Voisey's Bay), and 3) PGE-(Cu)-(Ni) deposits, which are enriched in PGE relative to Ni-Cu-Co (e.g., Bushveld, Great Dyke, Stillwater). Together, these deposits contain 56% of global Ni resources (the remainder in lateritic deposits) and 96% of global PGE resources (Mudd and Jowitt 2014; Barnes et al., 2017).

Giant Ni-Cu-PGE, Ni-Cu-(PGE), and PGE-(Cu)-(Ni) deposits/camps are distributed unevenly and significant deposits have been discovered in only a few countries (Australia, Canada, China, Russia, South Africa, USA) (Naldrett, 2004, 2010, 2011). Chinese resources in 2017 (USGS) were 2.5 Mt Ni (3.2% of global Ni resources) and 28 Mt Cu (3.9% of global Cu resources). Most Chinese deposits are relatively small (< 1 Mt Ni, except Jinchuan and Xiarihamu), have relatively low grades (< 1% Ni, except Hongqiling) (Table 1, Fig. 1), and are variably depleted in PGE relative to Ni-Cu-Co. Magmatic Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China account for 93% of total Ni resources (Sun et al., 2015), 6.4% of total Cu resources (Ying et al., 2014), and > 98% of total PGE resources (Zhou et al., 2002a).

Although the super-giant Noril'sk-Talnakh deposits formed at the Permian-Triassic boundary, most of the world's largest Ni-Cu-PGE, Ni-Cu-(PGE), and PGE-(Cu)-(Ni) deposits formed in the Neoarchean (e.g., Kambalda-Mt Keith-Perseverance, Great Dyke), Paleoproterozoic (e.g., Bushveld, Pechenga, Sudbury, Thompson), and Mesoproterozoic (e.g., Duluth, Voisey's Bay). The Jinchuan and Zhouan Ni-Cu-(PGE) deposits in China formed in the Neoproterozoic, but all other known Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China formed between the Early Devonian and Late Triassic (Table 1)

Like many Ni-Cu-PGE and Ni-Cu-(PGE) deposits worldwide, most Ni-Cu-(PGE) deposits in China are hosted by relatively small mafic-ultramafic intrusions. The Jinchuan Ni-Cu-(PGE) deposit, the largest single originally semi-continuous magmatic sulfide deposit in the world, occurs in a steeply-dipping intrusion with a surface cross-section of only 1.34 km^2 (Tang et al., 2009b, 2012b). Like some Ni-Cu-PGE deposits elsewhere in the world (e.g., Aguablanca, Ban Phuc Eagle, Expo-Méquillon, Santa Rita), many of the Ni-Cu-(PGE)-bearing intrusions in China have diamond-shaped surface sections and funnel-shaped cross sections (e.g., Huangshandong, Huangshannan, Huangshanxi, Jinchuan, Jingbulake, Kalatongke #1, Hongqiling #1, Limahe, Qingkuangshan, Zhubu) and have been interpreted to have filled subvertical transtensional features (Lightfoot and Evans-Lamswood, 2015).

The geology, mineralogy, petrology, geochemistry, and isotope geochemistry of Chinese Ni-Cu-(PGE) have been reported extensively in the literature (e.g., Zhou et al., 2002a, 2008; Song et al., 2009, 2012, 2016; Li et al., 2005, 2012; Wang et al., 2018). However, most papers describe single deposits or several deposits in a single region, leading to divergent interpretations regarding their genesis and metallogenesis. In particular, there is no interpretation of why Chinese Ni-Cu-(PGE) deposits as a whole are so different from those in other areas.

To aid in the interpretation of these deposits, we have compiled mineral chemical, whole-rock lithogeochemical, ore geochemical, and S-Nd-Sr-Os isotopic data from 18 representative Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China and use the data to generate a comparative view that provides better constraints on the genesis and metallogenesis of Chinese Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits.

2. Geological, Temporal, and spatial distribution

There are at least 80 magmatic Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China, including 2 giant deposits, 14 large deposits, 20 intermediate deposits, and many small deposits and occurrences (Lu et al., 2007). Some of the larger and more important deposits are shown in Fig. 2. Some appear to be isolated (e.g., Jinchuan), but many occur as clusters of individual deposits in the same area. For example, the Huangshan camp in the Eastern Tianshan district of the Xinjiang province includes several large- to medium-sized deposits, including Huangshandong, Huangshannan, and Huangshanxi, and the Pobei camp in the Beishan district of the Xinjiang province includes several cogenetic intrusions, including Poyi, Posan, Poqi, and Poshi. Bulk tonnages, bulk Ni grades, and contained Ni for Chines Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits are shown in Fig. 1. The resources for many deposits are not well characterized and many are low-grade deposits, but we include all for which we have data.

China is a collage of cratons and fragmental continental blocks and arcs, welded by ophiolitic mélange (Deng et al., 2014a,b, 2017). The principal tectonic provinces of China are shown in Fig. 2. The formation ages of the main magmatic sulfide deposits in China range from Neoproterozoic to Triassic with peaks in the Neoproterozoic and Permian (Fig. 3). The Neoproterozoic deposits occur along the margin of the North China Craton (e.g., Jinchuan) and the Yangtze Craton (e.g., Zhouan, Lengshuiqing). The Permian deposits occur mainly in the Central Asian Orogenic Belt (e.g., Huangshan, Tianyu, Kalatongke, Pobei) and the Emeishan Large Igneous Province (e.g., Baimazhai, Jinbaoshan, Limahe, Yangliuping, Zhubu). These locations represent rifted continental margins (e.g., Jinchuan), subduction and post-subduction of orogenic belts (e.g., Heishan, Huangshan, Kalatongke), and mantle plumes (e.g., Baimazhai, Jinbaoshan, Zhubu). Thus, based on mineralization age, tectonic setting, and spatial distribution, most deposits occur in 3 tectono-magmatic settings (Zhou et al., 2004; Lu et al., 2007; Sun et al., 2015): 1) Neoproterozoic deposits related to magmatic activity accompanying breakup of Rodinia (e.g., Jinchuan, Lengshuiqing, Zhouan); 2) Devonian to Triassic deposits related to subduction to post-collisional extension and magmatism in the orogenic settings (e.g., Chajianling, Erbutu, Jingbulake, Heishan, Hongqiling, Huangshan, Kalatongke, Lashuixia, Piaohechuan, Tianyu, Xiarihamu), and 3) Permian deposits related to plume magmatism in the Emeishan Large Igneous Province (e.g., Baimazhai, Jinbaoshan, Limahe, Yangliuping, Zhubu).

2.1. Neoproterozoic deposits related to breakup of Rodinia

The supercontinent Rodinia assembled 1.3–0.9 billion years ago and broke up 750–600 million years ago, triggered by an 830–820 Ma mantle plume beneath South China (Li et al., 1999, 2003b, 2008). According to Wang and Mo (1995), the North China, Tarim, and Yangtze Cratons appear to have been geographically closed during the assembly of the Rodinia supercontinent by ~ 1.0 Ga. The Ni-Cu-(PGE) deposits in the Baotan district, Guangxi province have been dated at 982 ± 21 Ma by whole-rock Re-Os n-TIMS methods, interpreted to represent an event that occurred during late assembly or early breakup of Rodinia (Mao and Du, 2001). The ~828 Ma Jinchuan and ~817 Ma Lengshuiqing Ni-Cu-(PGE) deposits have been interpreted to be related to a 830–820 mantle superplume accompanying the main phase of breakup of Rodinia (Li et al., 2005; Yang et al., 2005; Zhu et al., 2007; Zhang et al., 2010) and the ~637 Zhouan Ni-Cu-(PGE) deposit may represent the last magmatism related to the breakup of the Rodinia supercontinent at Neoproterozoic in the northern margin of the Yangtze Craton (Wang et al., 2013a).

The Jinchuan intrusion is located in the uplifted Longshoushan Terrane along the southwestern margin of the North China Craton. It intrudes Proterozoic (> 2 Ga) upper amphibolite facies orthogneisses, paragneisses, and marbles of the Baijiazhuizi Formation (Tang and Li, 1995; Lehmann et al., 2007; Tonnelier, 2010). There are four major groups of faults at Jinchuan: 1) steeply-dipping NW-SE striking regional thrust faults, 2) steeply-dipping NEE-SWW striking sinistral strike-slip faults, 3) NNW-SSE striking dextral strike-slip faults, and 4) steeply-dipping NE-SW striking normal faults, the second group of which

Table 1
Ages, mineralization types, tectonic settings, tectonic locations, morphologies, tonnages, grades, magma types, distances to craton margin, host rocks, associated rocks, and selected references for Chinese Ni-Cu-(PGE) and PGE-(Cu)-Ni deposits.

Deposit Name	Age (Ma)	Mineralization Type	Tectonic Setting	Tectonic Location	Morphology (surface section)	Morphology (cross section)	Surface Area (km^2)	Tonnes (Mt)	Ni Grade (%)	Cu Grade (%)
Jinchuan	826	Ni-Cu-(PGE)	Rifted margin	Rodinia	segmented elongated rhomb	narrow cone or sheet	1.4	550	1.05	0.7
Zhouan	637	Ni-Cu-(PGE)	Rifted margin	Rodinia	roughly triangular	elliptical	2.7	96	0.35	0.11–0.17
Jingbulake	431	Ni-Cu-(PGE)	Margin of orogenic belt	CAOB	elongate rhomb to curved cone that narrows into the structure at each end	not known	3	Not known	0.3–0.49	0.13
Xiarihamu	406	Ni-Cu-(PGE)	Orogenic belt	EKOB	ribbon-like	curved cone	0.8	157	0.64	0.14
Heishan	357	Ni-Cu-(PGE)	Margin of orogenic belt	CAOB	elliptical	curved cone	0.4	35	0.6	0.3
Kalatongke	287	Ni-Cu-(PGE)	Microcontinent	CAOB	weakly elongate rhomb	elliptical	0.25	30	0.88	1.3
Huangshannan	283	Ni-Cu-(PGE)	Along fault	CAOB	tube-like in east; lens-shaped in centre and west	curved cone or sills	4	30	0.4	0.1
Tianyu	280	Ni-Cu-(PGE)	Margin of terrane	CAOB	dyke-like	curved cone	0.2	8.3	0.6	0.24
Pobei	278	Ni-Cu-(PGE)	Margin of orogenic belt	CAOB	elliptical	curved cone	4	Not known	0.3–0.6	0.08–0.14
Huanshandong	270	Ni-Cu-(PGE)	Along fault	CAOB	elongate rhomb	curved cone that narrows into the structure at each end	5	135	0.3	0.16
Huanshanxi	269	Ni-Cu-(PGE)	Along fault	CAOB	ribbon-like	curved cone	3	80	0.54	0.3
Limahe	263	Ni-Cu-(PGE)	Intracontinental	ELIP	elongate rhomb	curved cone that narrows into the structure at each end with an asymmetric dyke-like keel	1.8	3	1	0.6
Yangliuping	263	Ni-Cu-(PGE)	Rifted margin	ELIP	sheet sill	sill	0.6	90	0.45	0.16
Zhubu	261	Ni-Cu-(PGE)	Rifted margin	ELIP	elliptical	curved cone	0.3	6.7	0.23	0.22
Jinbaoshan	260	PGE-(Cu)-(Ni)	Rifted margin	ELIP	sheet sill	sill-like	5	34	0.17	0.14
Baimazhai	259	Ni-Cu-(PGE)	Rifted margin	ELIP	U-tubed shape	curved cone	0.15	3.8	1.03	0.81
Hongqiling #7	216	Ni-Cu-(PGE)	Margin of orogenic belt	CAOB	sheet-like	curved cone	0.1	8.9	2.3	0.63
Piaohechuan	216	Ni-Cu-(PGE)	Margin of orogenic belt	CAOB	elongate rhomb	curved cone	0.07	1.2	0.83	0.05–0.39

Deposit Name	Magma	MgO (%)	FeOt (%)	Distance to craton margin (km)	Host Rock	Associated Rocks	References
Jinchuan	Ferropicrite; High Mg basalt	12.0–15.4	11.5–13.7	< 10	dunite, lherzolite	Proterozoic granitoids, gneisses, and marbles	Chai and Naldrett, 1992; Tonnelier 2010; Li and Ripley, 2011; Mao et al., 2018b
Zhouan	High Mg basalt	> 9.8		60	lherzolite	sandstone, conglomerate, and mudstone of the Zhujiashan Formation	Wang et al., 2012, 2013b
Jingbulake	High-Mg tholeiitic basalt	~12		60	Wehrlite, olivine gabbro, pyroxene diorite	gneiss and schist with lenses of marble	Yang and Zhou, 2009; Lightfoot and Evans-Lamswood, 2015
Xiarihamu	Boninitic basalt; High Mg basaltic magma	9.8			lherzolite, harzburgite, websterite, gabbro	Precambrian schist, granitic gneiss, and marble of the Proterozoic Jinshuikou Formation	Li et al., 2015; Song et al., 2016
Heishan	High Mg basalt	11.3	10.0	30	lherzolite, harzburgite	Neoproterozoic dolomitic marble and siliceous slate	Xie et al. 2012
Kalatongke	High Si and high Mg basaltic magma mixing	6.3–11.5	7.6		Diorite, Norite	Carboniferous Nanmingshu Formation clastic rocks	Zhang et al. 2009; Gao et al. 2012; Li et al. 2012
Huangshannan	Ni-rich magma; Picritic magma	8.7–12.4	9.1–10	120	lherzolite, websterite, gabbronorite, diorite	Lower Carboniferous quartz schist	Zhao et al., 2016a,b; Mao et al. 2016, 2017
Tianyu	High Mg basalt			70	Ol websterite, lherzolite	Precambrian metamorphic rocks with grades varying from greenschist to amphibolite	Tang et al., 2011, 2012a

(continued on next page)

Table 1 (*continued*)

Deposit Name	Magma	MgO (%)	FeOt (%)	Distance to craton margin (km)	Host Rock	Associated Rocks	References
Pobei	High Mg basalt	11.5–14.9	8.4–9.3	25	dunite,lherzolite, Ol websterite, gabbro, leucogabbro	Proterozoic schists, gneiss, and gneissic granite	Xia et al. 2013; Yang et al. 2014; Xue et al. 2016
Huanshandong	Basaltic magma	7.4–10	8.4–9.4	120	Lherzolite, Wehrlite, Gabbronorite	Lower Carboniferous Gandun Group turbidites	Sun et al. 2013; Mao et al. 2015
Huanshanxi	Tholeiitic basalt	8.7	8.4	120	Ol websterite, lherzolite, gabbro	Carboniferous Gandu Group schists	Zhang et al.2011, Mao et al. 2014
Limahe	Flood basalt/ Picritic magma			150	Wehrlite, Gabbro, Diorite	Neoproterozoic pyrite-bearing quartzites, graphitic slates, and siliceous limestones of the Huili Formation	Tao et al. 2008; Wang et al., 2018
Yangliuping	Flood basalt			90	Peridotite, Ol websterite, Ol clinopyroxenite, gabbro	Middle Devonian graphite-bearing schist, slate and graphite-bearing marble of the Weiguan Formation	Song et al.2003; Wang et al., 2018
Zhubu	Flood basalt/ Emeishan picrite			120	Lherzolite, Ol websterite, Gabbro, Gabbrodiorite	Precambrian gneiss of Yuanmou Formation	Tang et al. 2013
Jinbaoshan	Flood basalt/ Picritic magma	11.0	14.0	< 10	Wehrlite	Devonian dolomite, sandstone and slate of the Jinbaoshan Formation	Tao et al. 2007; Wang et al. 2010
Baimazhai	Flood basalt/ Mg-rich picritic magma			20	websterite, gabbro	Ordovician sandstone and slate of the Xiangyang Formation	Wang and Zhou,2006; Wang et al. 2006
Hongqiling #7	High Mg basalt			< 10	Orthopyroxenite, harzburgite, norite	Hulan Group gneiss, schist and marble	Wei et al. 2013
Piaohechuan	High Mg basalt			< 10	Horblende olvine gabbro, Horblende gabbro	Hulan Group gneiss, schist and marble	Wei et al. 2015

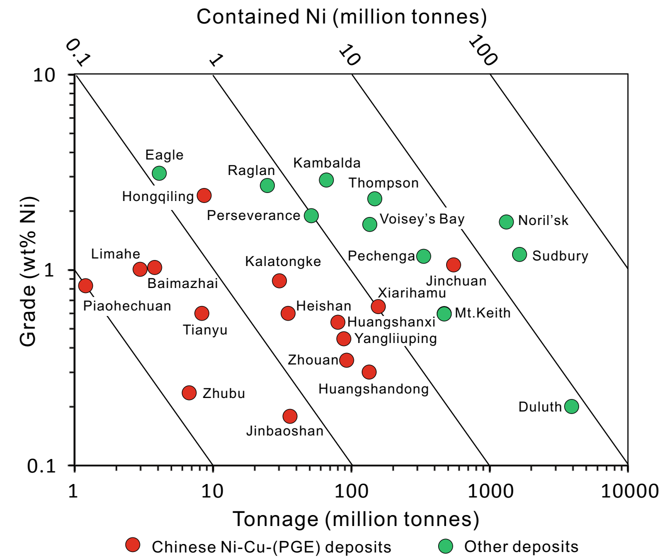

Fig. 1. Plot of Ni grade vs. tonnage of deposit for Chinese Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits and other Ni-Cu-PGE deposits worldwide. Data sources: Naldrett (2004), Wang and Zhou (2006), Lu et al., (2007), Tao et al. (2008), Song and Li (2009), Ding et al. (2012), Zhang et al. (2011), Tang et al. (2012a), Gao et al. (2013), Xie et al. (2014), Wei et al. (2015), and Song et al. (2016).

divides the intrusion into 4 segments (from west to east): Mining Areas III, I, II, and IV (Tang, 1993; Tang and Li, 1995; Song et al., 2012). The predominant rock types in the Jinchuan intrusion are heteradcumulate olivine-sulfide (net-textured mineralization), mesocumulate lherzolite, orthocumulate olivine websterite, and minor plagioclase lherzolite and gabbro (Tonnelier, 2010; Duan et al., 2016). Zircons from a dolerite dike cutting the intrusion and a lherzolite sample from the intrusion have been dated at 828 ± 3 Ma and 831 ± 0.6 Ma, respectively by U-Pb SHRIMP and U-Pb ID-TIMS single zircon methods, suggesting that the emplacement of the Jinchuan intrusion was related to the breakup of Rodinia (Li et al., 2005; Zhang et al., 2010).

The Zhouan intrusion is located along the northern margin of the Yangtze Block (Wang et al., 2013a,b). Is it one of numerous mafic–ultramafic bodies occurring in the Southern Qinling orogenic belt and along margin of Yangtze Craton, which range 820–632 Ma in age and which are considered to have formed from a magma derived from upwelling asthenosphere mantle during Neoproterozoic extension related to breakup of Rodinia in that region (Gao et al., 1992, 1996; Li et al., 2003b; Zeng et al., 2016). The Zhouan intrusion, dated at 637 ± 4 Ma and associated with several other ~635 Ma mafic–ultramafic intrusions in the Suizao basin, appears to represents some of the last magmatism related to the breakup of Rodinia along the northern margin of the Yangtze Craton (Wang et al., 2013a). The intrusion is a 2450 m long sill-like body that is 700 m wide in the western part and 1500 m wide in the eastern part. It consists of lherzolite and olivine gabbronorite and intrudes the Meso- to Neoproterozoic Zhujiashan Formation, which is overlain by 30–120 m thick Cenozoic sedimentary rocks (Wang et al., 2013b).

2.2. *Permian to Triassic deposits in the central Asian orogenic belt (CAOB) and an early Devonian deposit in the East Kunlun orogenic belt*

The Central Asian Orogenic Belt (CAOB) extends over 5000 km, from the Urals Mountains in Russia in the west through Kazakhstan, Uzbekistan, Tajikistan, Kyrgyzstan, the Xinjiang province in northwestern China, and parts of Mongolia and Inner Mongolia to Northeast China (Fig. 2). It is a complex collage containing continental fragments, island arc assemblages, remnants of oceanic crust, and continental margins that is sandwiched between the Siberian craton to the north

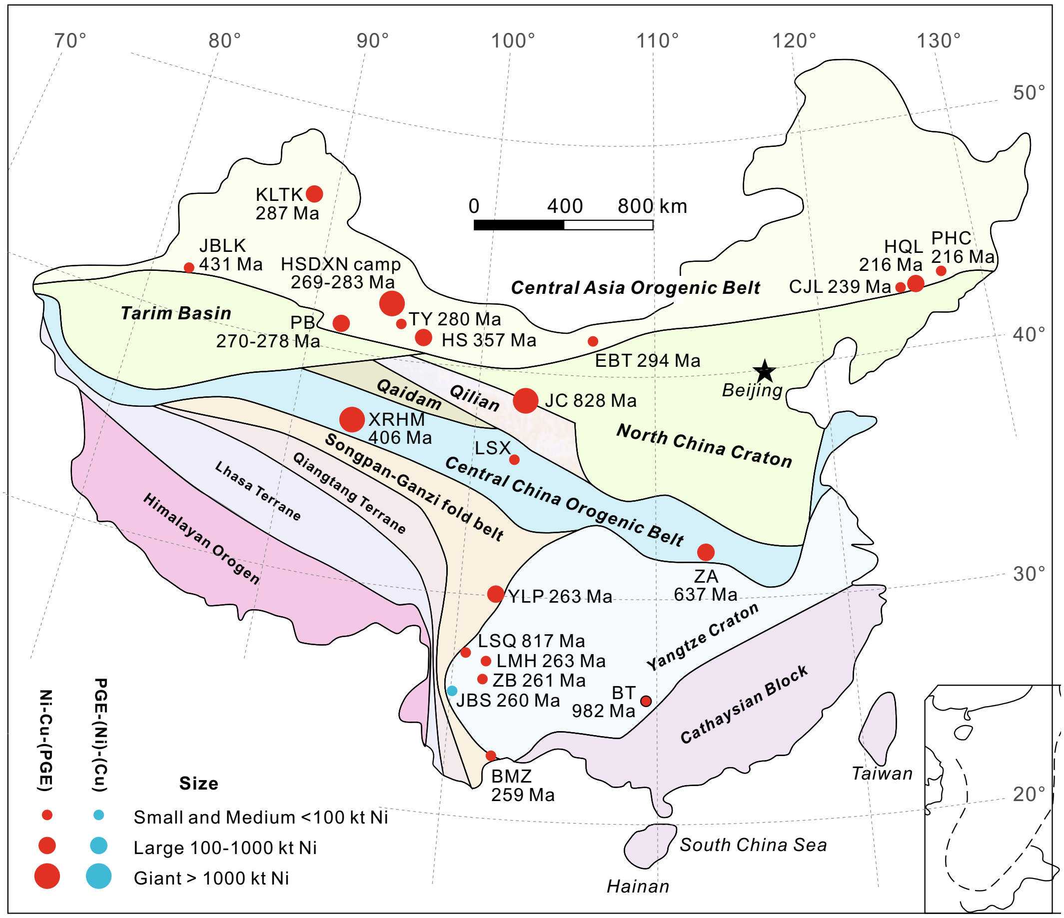

Fig. 2. Simplified geological map of continental China showing major terrains and shield areas, the locations of magmatic sulfide deposits, and the relative sizes of their Ni resources. The ages of dated deposits are from Han et al. (2004), Wu et al. (2004), Zhou et al. (2004), Li et al. (2005), Wang et al. (2006), Zhou et al. (2008), Tao et al. (2009), Yang and Zhou (2009), Zi et al. (2010), Tang et al. (2011), Zhang et al. (2011), Xie et al. (2012), Zhang (2013), Hao et al. (2013), Wang et al. (2013a), Peng et al. (2013), Xia et al. (2013), Zhao et al. (2015), and Song et al. (2016). BMZ Baimazhai, BT Baotan; CJL Chajianling; EBT Erbutu; HS Heishan; HSDXN camp Huangshandong, Huangshanxi, and Huangshannan; HQL Hongqiling; JBS Jinbaoshan; JC Jinchuan; JBLK-Jingbulake; KLTK Kalatongke; LMH Limahe; LSX Lashuixia; PHC Piaohechuan; PB Pobei; TY-Tianyu; XRHM Xiarihamu; YLP Yangliuping; ZB Zhubu; ZA Zhouan.

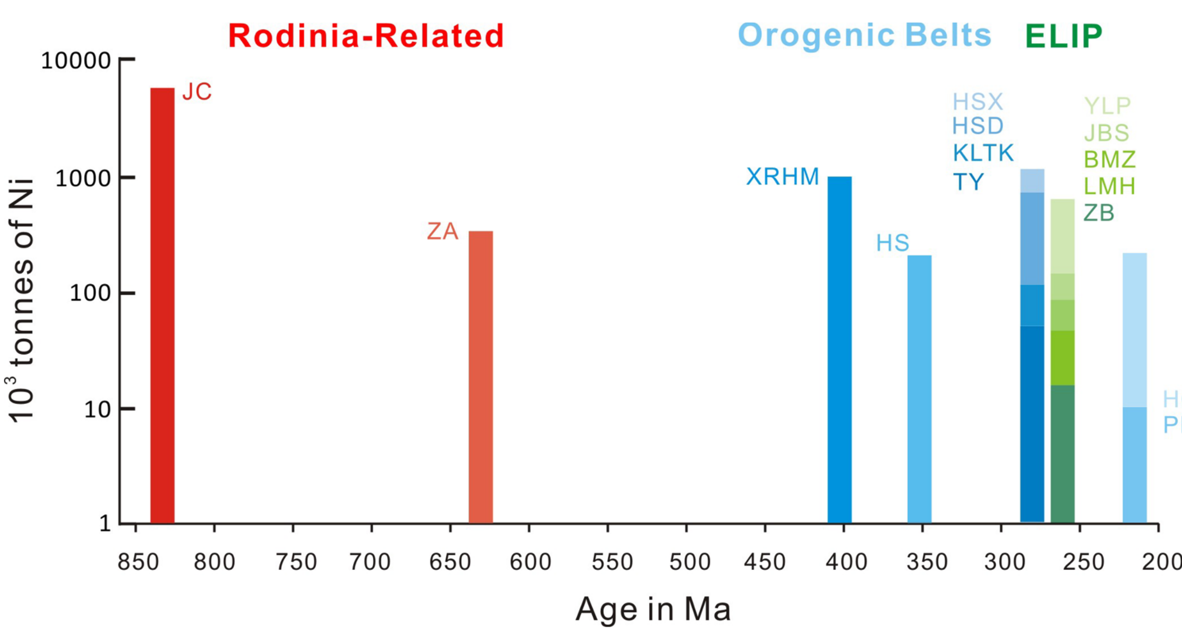

Fig. 3. Ages of Chinese magmatic sulfide deposits. Data sources in Fig. 1.

and the Tarim-North China craton to the south. It appears to have formed by multiple subduction-accretion and collision processes between the Neoproterozoic and Late Paleozoic, and contains considerable amounts of Cu, Au, Pb-Zn, and Ni-Cu-(PGE) mineralization (Sengör et al., 1993; Xiao et al., 2003, 2008, 2009; Windley et al., 2007; Xie et al., 2014; Deng and Wang, 2016; Mao et al., 2018a). The CAOB in China can be subdivided into a western Tianshan-Altay segment (Mao et al., 2008) and an eastern segment in Inner Mongolia and northeastern China (Zhou and Wilde, 2013). Both parts contain numerous mafic–ultramafic intrusions, most of which appear to have formed in the late Permian to Triassic (e.g., Huangshan, Kalatongke, Pobei, Tianyu, Hongqiling, Piaohechuan), although some intrusions and minor magmatic sulfide deposits formed in the Silurian and Carboniferous, including the ~431 Ma Jingbulake Ni-Cu-(PGE) deposit and the ~357 Ma Heishan Ni-Cu-(PGE) deposit (Yang and Zhou, 2009; Xie et al., 2014).

The western part of the CAOB can be subdivided into 4 areas from north to south. 1) The Altay orogenic belt is bounded to the south by the Ulungur fault and the Junggar block and to the north by the Sayan and associated belts, and has been interpreted to represent a late Cambrian to Carboniferous magmatic arc built upon a microcontinent (Sengör et al., 1993; Windley et al., 2002; Xiao et al., 2009). The ~287 Ma Kalatongke Ni-Cu-(PGE) deposit is located in this belt, hosted in a mafic intrusion consisting of norite, troctolite, gabbro, and diorite (Han et al., 2004; Gao et al., 2012). 2) The Eastern Tianshan district is located between the Junggar block and Tarim block and contains ~120 mafic–ultramafic intrusions, at least six of which host significant Ni-Cu-(PGE) mineralization, including the Huangshandong, Huangshanxi, Huangshannan, Hulu, Tulaergen, and Xiangshan deposits (Zhou et al., 2004; Mao et al., 2008; Qin et al., 2011; Gao et al, 2012; Tang et al., 2012a). It is considered to be the most important Ni-Cu-(PGE) belt in northwestern China and great success has been achieved recently in the exploration for additional mineralization in this district (Mao et al., 2008; Han et al., 2010; Tang et al., 2012a). 3) The Central Tianshan district is located between the Kawabulak fault, which marks the northern boundary of the South Tianshan orogen and the Aqikkuduk fault, which marks the southern boundary of the Jueluotage orogenic belt (Chai et al., 2008; Song et al., 2013). Several mafic–ultramafic intrusions occur on the southern side of the Shaquanzi fault, some of which contain Ni-Cu-(PGE) mineralization, including the ~280 Ma Tianyu and ~281 Ma Baishiquan deposits (Tang et al., 2011; Mao et al., 2006). 4) The Beishan district is an E-W trending mountain range located along the northeastern margin of Tarim Craton, associated with several mafic–ultramafic intrusions containing significant Ni-Cu-(PGE) mineralization. The Pobei complex is the largest and is composed of a gabbroic unit and two ultramafic bodies (Poshi and Poyi). Zircon U-Pb dates range 270–278 Ma and it is considered to form by multiple pulses of olivine and sulfide-charged magma (Li et al., 2006; Yang et al., 2014; Xue et al., 2016).

The Hongqiling-Chajianling-Piaohechuan district in the most eastern part of the CAOB contains nearly 200 mafic–ultramafic intrusions, many of which have been dated between ~216 and ~239 Ma and contain Ni-Cu-(PGE) mineralization (Wu et al., 2004; Hao et al., 2014; Wei et al., 2015). The ~216 Ma Hongqiling Ni-Cu-(PGE) deposit, the largest in this district and prior to 2006 the second largest producer of Ni after Jinchuan (Lu et al., 2011; Wei et al., 2013), is located in the southeastern part of the CAOB adjacent to the North China Craton, and is composed mainly of orthopyroxenite, harzburgite, and norite (Wei et al., 2013). The ~216 Ma Piaohechuan Ni-Cu-(PGE) deposit occurs 80 km northeast of Hongqiling (Lu et al., 2007) and is hosted by a very small (~0.07 km^2) intrusion composed of hornblende-olivine gabbro and hornblende gabbro (Wei et al., 2015).

The 431–405 Ma (Li et al., 2015; Peng et al., 2016; Song et al., 2016) Xiarihamu Ni-Cu-(PGE) deposit was discovered in 2011 by the Qinghai Bureau of Geology and Mineral Resources during a regional exploration program for hydrothermal base and precious metal mineralization in the East Kunlun orogenic belt. The East Kunlun orogenic belt merges with the Qinling-Dabie orogenic belt in the east, separating the North China Craton and South China Block (Li et al., 2015; Yang et al., 2015a,b). The Xiarihamu deposit is not only the second largest Ni-Cu deposit in China (after Jinchuan), but is also the largest magmatic Ni-Cu ± PGE deposit discovered thus far in a possible arc setting (Li et al., 2015; Song et al., 2016). The host intrusion is ~500 m wide and 1600 m long, and composed of lherzolite and olivine websterite with minor dunite, websterite, and orthopyroxenite. Mineralization occurs mainly as disseminated to net-textured zones within the ultramafic body and near the contact with older gabbro (Li et al., 2015; Zhang et al., 2017). The age suggests a genetic linkage with Early Devonian regional magmatism (Song et al., 2016).

2.3. Permian deposits in the Emeishan large Igneous province

The ~260 Ma Emeishan Large Igneous Province (ELIP) is located along the western margin of the Yangtze craton in southern China and is composed of voluminous flood basalts and associated mafic–ultramafic intrusions extending over an area of ~3×10^5 km^2 (Xu et al., 2001, 2004; Zhou et al., 2002b, 2008; Deng et al., 2010; Wang et al., 2007a,b, 2011; Li et al., 2016). It is bounded to the west by the Ailaoshan-Red River (ASRR) fault and to the northwest by the Longmenshan-Xiaojinghe fault (Xu et al., 2001). ELIP basalts can be geochemically subdivided into low-Ti and high-Ti types based on Ti contents and Ti/Y ratios (Xu et al., 2001; Xiao et al., 2004). The high-Ti basalts have been linked to Fe-Ti-V oxide deposits (e.g., Panzhihua), whereas the low-Ti basalts have been linked to Ni-Cu-(PGE) magmatic sulfide deposits (Zhou et al., 2008; Wang et al., 2011, 2018). The ELIP hosts the widely variety of magmatic sulfide deposits in China, including Ni-Cu-(PGE) deposits (e.g., Baimazhai, Limahe, Qingkuangshan, Yangliuping, Zhubu) and one PGE-(Cu)-(Ni) deposit (e.g., Jinbaoshan) (Zhou et al. 2002b; Song et al., 2003; Wang and Zhou, 2006; Tao et al., 2008; Song et al., 2008; Wang et al., 2010; Zhu et al., 2011; Lu et al., 2014).

The ~260 Ma Jinbaoshan PGE-(Cu)-(Ni) deposit is located in the central zone of the ELIP and is the largest PGE-(Ni)-(Cu) deposit in China (Table 1). It is hosted by a sill-like ultramafic intrusion that is ~5 km long, up to 1.2 km wide, and up to 170 m thick that is composed mainly of wehrlite (92%) with minor gabbro that intrudes Devonian dolomite (Tao et al., 2007; Wang et al., 2010). The mineralization is dominated by a violarite(polydymite)-pyrite-millerite-chalcopyrite mineral assemblages and has been referred to in the literature as a Pt-Pd deposit, but it contains proportional amounts of other PGE, including Ir, which has important genetic implications (Lu et al., *in prep*.)

The Ni-Cu-(PGE)-bearing ~261 Ma Zhubu Ni-Cu-PGE intrusion, also located in the central zone, is composed of lherzolite, olivine websterite, and contaminated gabbroic rocks (Zhou et al., 2008; Tang et al., 2013). The sulfide mineralization occurs mainly as disseminated pyrrhotite, pentlandite, and chalcopyrite within the marginal zone (Tang et al., 2013).

The Yangliuping Ni-Cu-(PGE) camp is located in southeastern part of the Mesozoic Songpan-Ganzi orogenic belt. The currently known economic deposits occur in the Yangliuping and Zhengziyanwo sills, which are 2–3 km long, up to 300 m thick, extend 300–500 m along the dip, and are composed by serpentinite, talc schist, tremolite schist, and metagabbro (Song et al., 2003). The sulfides in Yangliuping vary from massive sulfides with up to ~34% S to disseminated sulfides with as little as 1% S.

The ~263 Ma Limahe Ni-Cu-(PGE) deposit is located in the central zone. The surface exposure of the intrusion is about 900 m long and 180 m wide (Zhou et al., 2008; Tao et al., 2008). It is a multiphase intrusion with an ultramafic (wehrlite and olivine gabbro) lower part and a mafic (diorite and gabbro) upper part (Tao et al., 2008).

The ~259 Ma Baimazhai Ni-Cu-(PGE) deposit is located in the

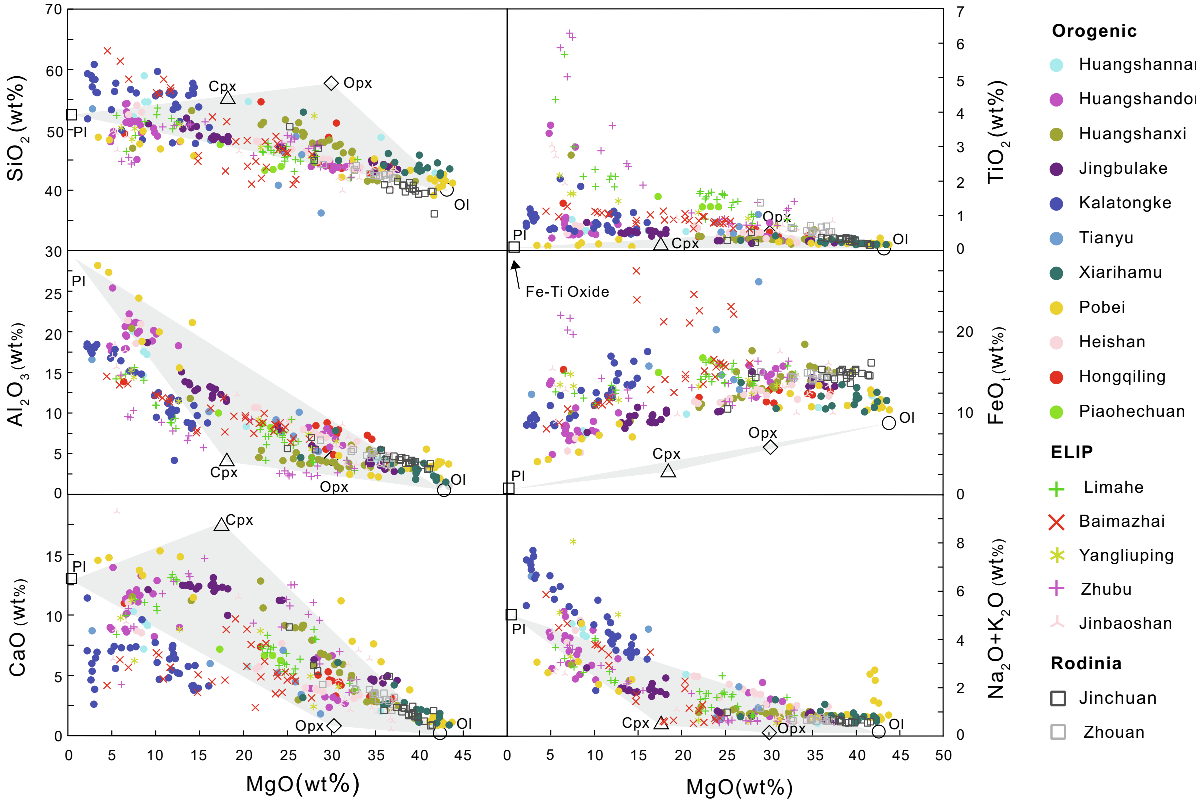

Fig. 4. Whole-rock SiO_2, TiO_2, Al_2O_3, FeO_t, CaO, and $Na_2O + K_2O$ vs MgO. Data from Song et al. (2003), Wang et al. (2006), Tao et al. (2007, 2008), Xie et al. (2007), Yang and Zhou (2009), Tang et al. (2011), Li et al. (2012, 2015), Xie et al. (2012), Song et al. (2012), Sun et al. (2013), Tang et al. (2013), Xia et al. (2013), Wang et al. (2013b), Mao et al. (2014), Zhao et al. (2015), and Li et al. (2018). Olivine, orthopyroxene, clinopyroxene, and plagioclase data are average compositions at Jinchuan (Tonnelier, 2010). Samples with Ni + Cu > 0.5% have been excluded.

Jinping area, on the southwest margin of the Yangtze Plate, near the Ailaoshan-Red River (ASRR) fault zone. The deposit is hosted in a lens-shaped mafic–ultramafic body that is ~530 m long, ~190 m wide, up to 64 m thick, and intrudes Ordovician metasandstone and slate. It contains significant amounts of Ni-Cu-(PGE), occurring as massive to net-textured sulfides in a zoned orthopyroxenite-websterite-gabbro body (Wang and Zhou, 2006).

3. Geochemistry

3.1. Lithophile geochemistry

Major and trace elements for samples from 18 representative mineralized intrusions for which high-quality data are available have been recalculated to 100% on an anhydrous basis and plotted in Fig. 4. Only samples with Ni + Cu ≤ 0.5 wt% have been included, filtering out samples containing significant amounts of Fe-Ni-Cu sulfides. The low-sulfide host rocks include dunites, lherzolites, websterites, wehrlites, gabbros, and norites, exhibiting a wide range of MgO (up to 44%) and SiO_2 (up to 61%). To aid in evaluating the broader mineralogical controls, we have also plotted the average compositions of olivine, orthopyroxene, clinopyroxene, and plagioclase at Jinchuan from Tonnelier (2010). Most sulfide-poor samples fall in the field outlined by the compositions Ol-Opx-Cpx-Pl, consistent with their observed cumulus mineralogy. Samples from Pobei have very high $Na_2O + K_2O$ contents because they contain hornblende and plagioclase (Su et al., 2011). On the TiO_2 and FeO_t vs MgO, many sulfide-poor samples plot outside the Ol-Opx-Cpx-Pl field, because of the presence of Fe ± Ti ± Cr oxides (magnetite, ilmenite, chromite).

The Ni-Co-Cr contents of sulfide-poor samples decrease with decreasing Mg (Fig. 5), consistent with fractional crystallization/accumulation of olivine and minor chromite, except for several samples at Xiarihamu, which show a trend consistent with fractional crystallization/accumulation of only olivine (see discussion by Lesher and Stone, 1996; Barnes, 1998). Jinbaoshan samples are enriched in Cr because they contain cumulus chromite.

Highly incompatible lithophile elements (HILE: U-Th-Nb-Ta-Zr-Hf-REE-Y) generally increase with decreasing Mg (Fig. 6), reflecting fractional crystallization/accumulation of olivine ± Opx. Kalatongke and localities in the ELIP have higher La and La/Sm, and lower Ba/Nb than other deposits in the CAOB or associated with Rodinia. Some of the variation in Ba/Nb may reflect mobility of Ba, but some appears to be magmatic. Localities in the ELIP other than Baimazhai have somewhat higher Gd/Yb ratios. Deposits associated with the breakup of Rodinia, especially Jinchuan, exhibit quite variable Ba/Nb and La/Sm, some of which may reflect mobility of Ba and La, but much of which appears to be magmatic.

Sr and Nd isotopic compositions are shown in Fig. 7. Huangshannan, Huangshandong, and Kalatongke have $\varepsilon Nd_{(t)}$ values close to MORB, suggesting minor crustal contamination or significant amounts of magma replenishment (see discussion Lesher and Arndt, 1995; Lesher and Burnham, 2001). Heishan, Hongqiling, Jingbulake, Piaohechuan, Pobei, and Tianyu have slightly lower $\varepsilon Nd_{(t)}$ values, consistent with some crustal contributions. Host rocks in the CAOB (Heishan, Huangshandong, Huangshangnan, Hongqiling, Jingbulake, Pobei, Tianyu) generally have higher $\varepsilon Nd_{(t)}$ and lower $^{87}Sr/^{86}Sr_{(i)}$, consistent with mantle sources, whereas host rocks in the ELIP (Baimazhai, Jinbaoshan, Limahe, Zhubu) have lower $\varepsilon Nd_{(t)}$ and higher $^{87}Sr/^{86}Sr_{(i)}$ values, defining a trend from ELIP flood basalts towards upper continental crust. Deposits associated with the break-up of Rodinia (Jinchuan and Zhouan) also have lower $\varepsilon Nd_{(t)}$, consistent with an enriched mantle source and/or crustal contamination, but show a different trend to the

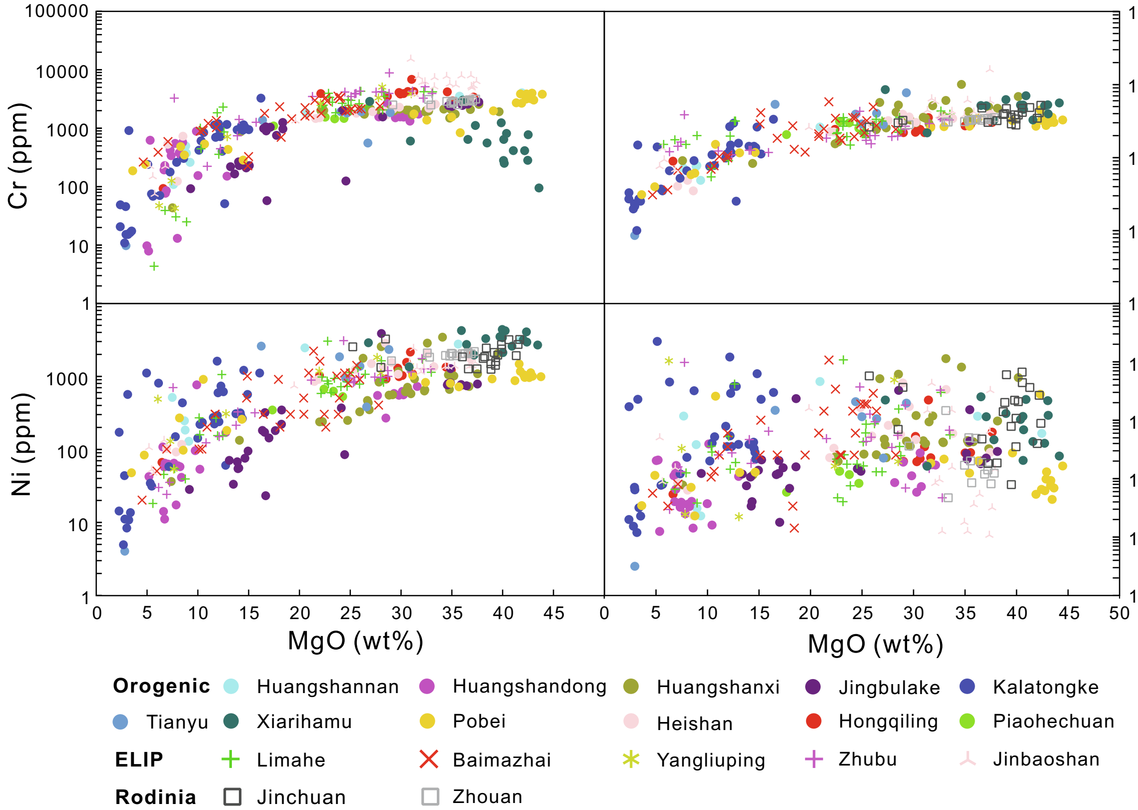

Fig. 5. Whole-rock Cr, Co, Ni, and Cu vs MgO. Data sources as in Fig. 4.

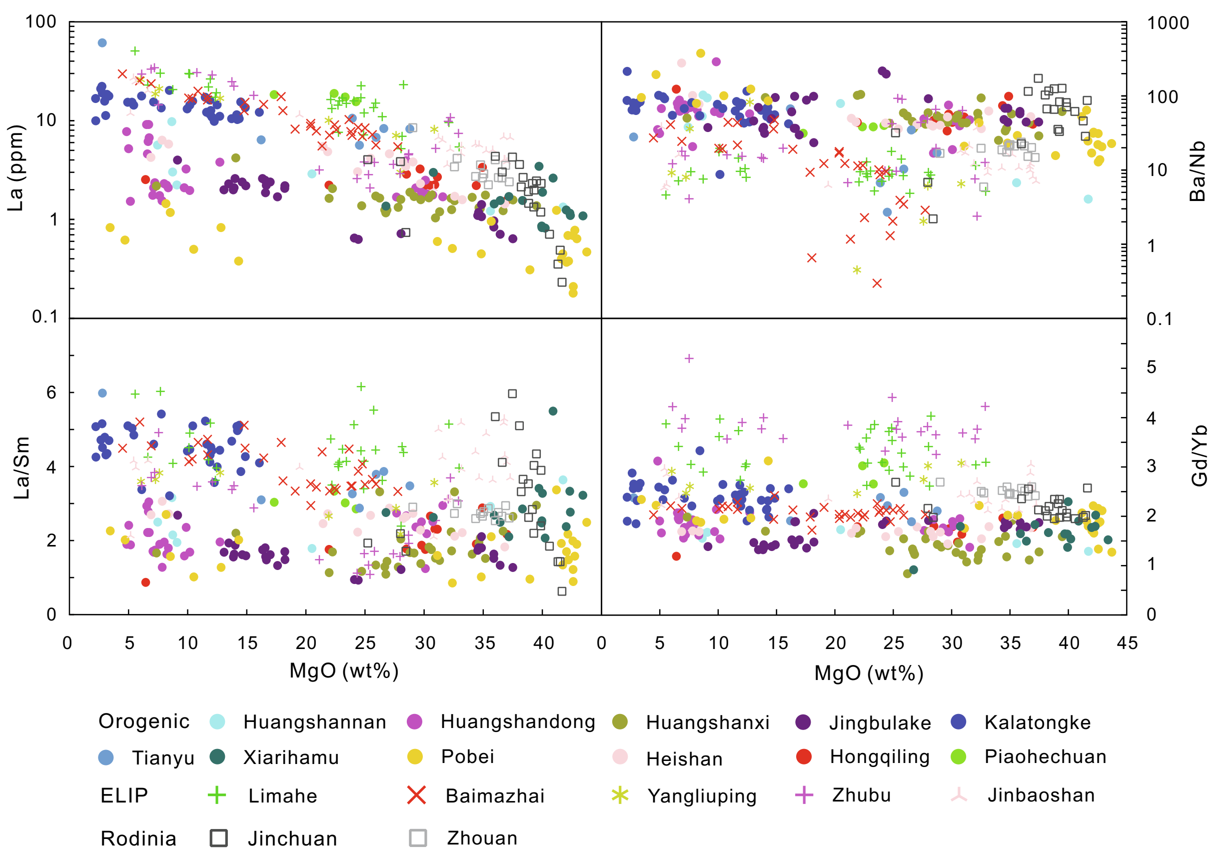

Fig. 6. Whole-rock La, Ba/Nb, La/Sm, and Gd/Yb vs MgO. Data sources as in Fig. 4.

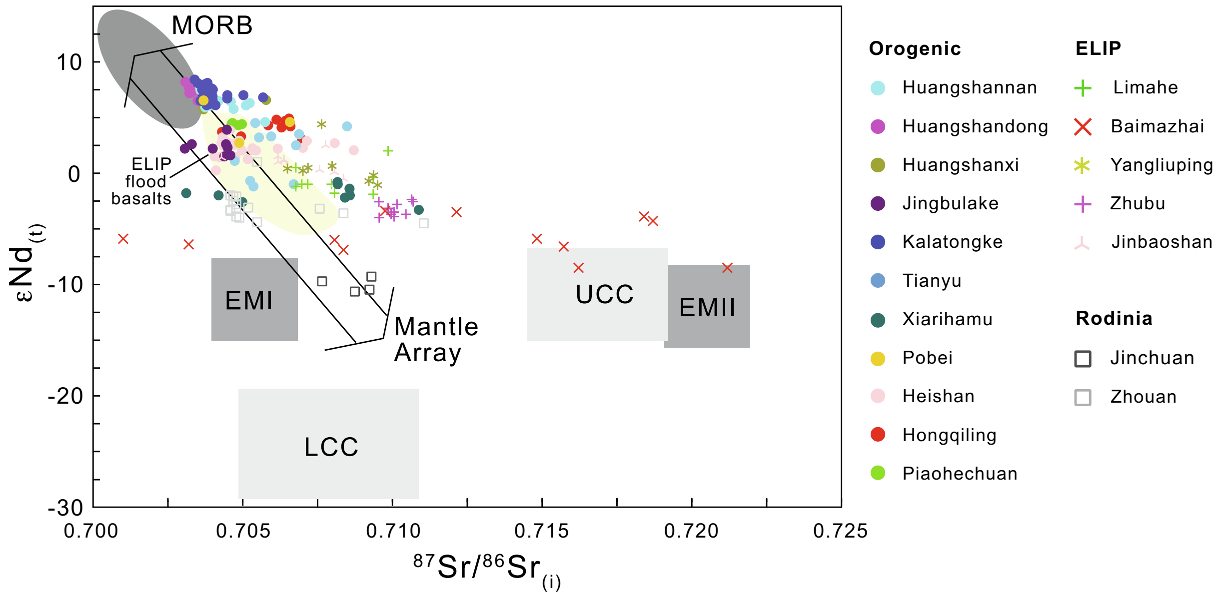

Fig. 7. $\varepsilon Nd_{(t)}$ versus $^{87}Sr/^{86}Sr_{(i)}$ for Chinese Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits. Data sources: magmatic sulfide deposits from Wang et al. (2006), Zhou et al. (2008), Song and Li (2009), Tang et al. (2011), Yang et al. (2012), Xia et al. (2013), Tang et al. (2013), Sun et al. (2013), Wang et al. (2013b), Wei et al. (2013, 2015), Duan et al. (2016), and Peng et al. (2016); MORB, EMI, EMII, and mantle array from Zindler and Hart (1986); LCC and UCC from Jahn et al. (1999).

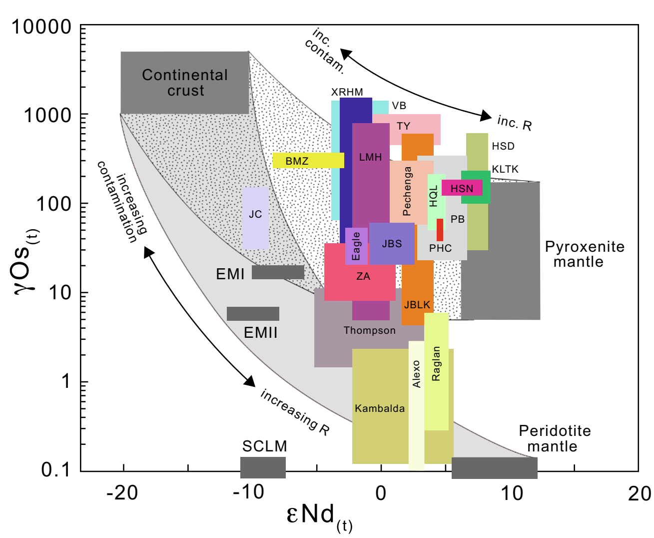

Fig. 8. εNd vs γOs for Chinese Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits. Data sources: Nd isotopes of Chinese deposits from references in Fig. 7; Os isotopes of Chinese deposits from Yang et al. (2005), Wang et al. (2007a,b), Han et al. (2007), Zhang et al. (2008), Sun et al. (2008), Tao et al. (2010a, 2010b), Tang et al. (2011), Yang et al. (2008, 2012, 2014), Gao et al. (2013), Wang et al. (2013b), Wei et al. (2013, 2015), Ling (2014), Zhang et al. (2017), Mao et al. (2017). Eagle from Ding et al. (2012); Pechenga from Walker et al. (1994) and Hanski et al. (2014); Voisey's Bay from Amelin et al. (2000) and Lambert et al. (2000); Kambalda and Thompson from Lesher and Arndt (1995), Foster et al. (1996), Zwanzig (2005), and Hulbert et al. (2005); Alexo and Raglan from Shirey and Barnes (1994), Gangopadhyay and Walker (2003), and Dupré et al. (1984); peridotite mantle from Rudnick et al. (2004); pyroxenite mantle from Lassiter et al. (2000) and Pearson and Nowell (2004); EMIand EMIIfrom Zindler and Hart (1986) and Shirey and Walker (1998); SCLM from Ding et al. (2012); continental crust from Jahn et al. (1999) and Lambert et al. (1999).

upper continental crust. These data provide clear evidence for variable degrees of crustal contamination and indicate that the Nd isotopic composition of the crust sampled by the host magmas was different in different areas, as would be expected.

εNd and γOs data are plotted in Fig. 8. Most samples from Chinese deposits except Jinchuan have high γOs and intermediate εNd, plotting on a mixing trend between magmas derived from pyroxenitic mantle and continental crust, similar to Voisey's Bay and Eagle, but distinct for the most part from komatiite-associated deposits, which have low γOs and intermediate εNd, plotting on a mixing trend between magmas derived from peridotitic mantle and continental crust. All are distinctly different from subcontinental lithospheric mantle (SCLM), which has lower εNd and lower γOs reflecting its depleted nature, and OIB sources (EMI, EMII), which also have low εNd but higher γOs reflecting their enriched nature. Samples from Jinchuan plot at lower εNd values but equally high γOs values, plotting on a mixing trend between continental crust and magmas derived from either peridotitic or pyroxenitic mantle (or a hybrid). Deposits in the CAOB plot closer to pyroxenitic mantle than deposits in ELIP and deposits related to Rodinia breakup. γOs values are typically more variable than εNd values, reflecting the higher Os contents of sulfides: removing sulfides will increase Re/Os, evolving over time to higher $^{187}Os/^{186}Os$ and γOs, whereas adding sulfides will decrease Re/Os, evolving over time to lower $^{187}Os/^{186}Os$ and γOs. Overall, deposits in the CAOB exhibit higher εNd and lower εSr (i.e., lower $^{87}Sr/^{86}Sr_{(i)}$), and therefore appear to be less contaminated than those in the ELIP, but there are no systematic differences in γOs.

3.2. Metal geochemistry

Ni, Cu, Co, and PGE in mineralized samples correlate positively with S in most deposits (Fig. 9), suggesting that these elements were originally housed in sulfide melts. However, the PGE contents vary more widely than Ni-Co-Cu, likely reflecting differences in magma:sulfide ratio (R factor: Campbell and Naldrett, 1979) and some sulfide-rich samples (e.g., Baimazhai) show negative correlations, reflecting accumulation of MSS (see below). Among the PGEs, Pt generally shows weaker correlations and some deposits (e.g., Huangshandong, Piaohechuan, Xiarihamu) show more scatter, which is also obvious in their primitive mantle-normalized PGE patterns (Fig. 10) and may reflect accumulation of MSS or residual sulfide liquid and/or "nugget" effects.

In order to compare samples with different sulfide contents, we have recalculated the metal abundances to 100% sulfide (Naldrett, 1981; Barnes and Lightfoot, 2005):

$$C^{i}_{100} = 100\,C^{i}_{WR}/(2.527\ S + 0.3408\ Cu + 0.4715 Ni) \quad (1)$$

where C^{i}_{100} = concentration of element i in 100% sulfides,

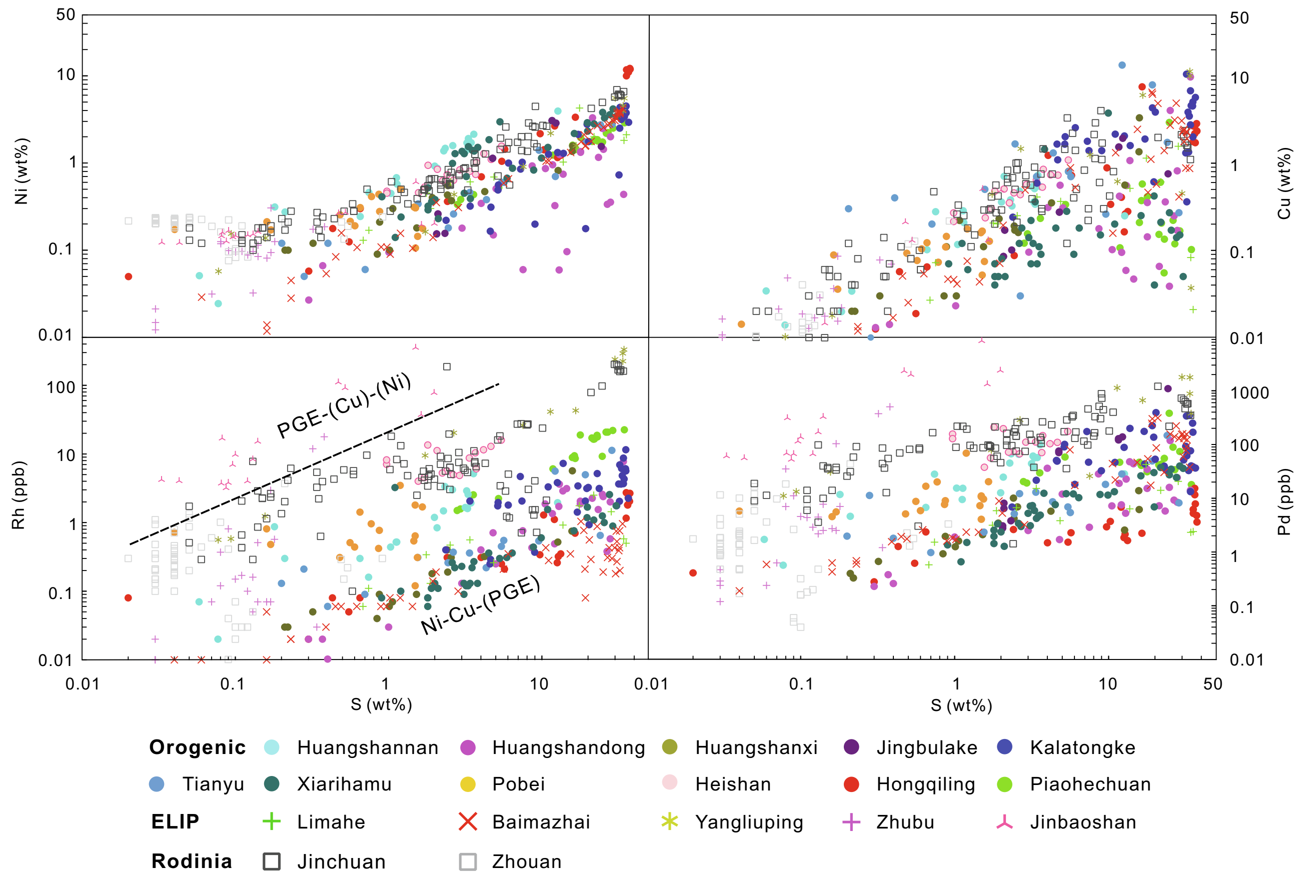

Fig. 9. Whole-rock Ni, Cu, Rh, and Pd vs S. Data sources: Wang et al. (2006), Wang and Zhou (2006), Tao et al. (2007, 2008), Song et al. (2008, 2016), Tang et al. (2011), Yang et al. (2012, 2014), Li et al. (2012), Tang et al. (2013), Wei et al. (2013); Gao et al. (2013), Wang et al. (2013b), Chen et al. (2013), Mao et al. (2014), Xie et al. (2014), Zhao et al. (2015), Wei et al. (2015).

C^{i}_{WR} = concentration of the element in the whole rock, and S, Cu, and Ni = concentrations of those elements in the whole rock in wt%. This calculation assumes that all of the Cu is in stoichiometric chalcopyrite ($CuFeS_2$), that all of the Ni is in pentlandite containing 34% Ni, and that the remaining S is in pyrrhotite containing 63% Fe, and it does not include a correction for silicate Ni. The Ni contents of pentlandite and Fe contents of pyrrhotite will obviously vary from deposit to deposit and sample to sample, but this calculation is sufficient for the purposes of comparison. Because the Jinbaoshan PGE deposit contains mainly violarite, chalcopyrite, and pyrite, we assume that all of Cu is in chalcopyrite, all of the Ni is in violarite, and the remaining S is in pyrite.

PGE_{100} values generally correlate positively between Ir and other PGEs (Fig. 11), although parts of some deposits show negative correlations (e.g., Baimazhai, Jinchuan, and Piaohechuan), confirming accumulation of MSS or residual sulfide liquid.

The abundances of chalcophile elements in 100% sulfides have been normalized to primitive mantle (Barnes and Maier, 1999) and plotted in terms of increasing compatibility during mantle melting in Figs. 10–11. Baimazhai, Hongqiling, Huangshandong, Huangshannan, Huangshanxi, Jingbulake, Kalatongke, Limahe, Tianyu, and Xiarihamu are more enriched in Ni-Cu relative to PGE, whereas Heishan, Pobei, Yangliuping, and Piaohechuan are less enriched in Ni-Cu relative to PGE. Jinchuan, Zhouan, and Zhubu exhibit a range of patterns, but include some that are enriched in Ni-Cu relative to PGE. Jinbaoshan is enriched in Pd-Pt-Rh-Ir relative to Ru and Cu-Ni, similar to other PGE deposits (e.g., Bushveld, Great Dyke, Stillwater).

S isotopic compositions of mineralized samples are shown in Fig. 12. Most (e.g., Hongqiling, Huangshannan, Huangshandong, Jinbaoshan, Jinchuan, Kalatongke, Pobei, and Yangliuping) range between −2 to 2‰ $\delta^{34}S$ and are close to – but in many cases significantly different from – the 0.1 ± 0.5‰ value expected for high-degree melting of asthenospheric mantle (Lesher, 2017) and therefore require some contributions by crustal S, which may have exchange with magmatic S at moderate-high magma:sulfide ratios (see Lesher and Stone, 1996; Lesher and Burnham, 2001). Some (e.g., Baimazhai, Limahe, and Xiarihamu) have more positive $\delta^{34}S$ values, even more clearly requiring crustal S. The No. 1 body at Heishan ranges 0.43–1.0‰ $\delta^{34}S$, whereas the No. 4 body at Heishan ranges 1.9–6.1‰ $\delta^{34}S$, indicating different S sources or different magma:sulfide ratios (higher in No. 1 and lower in No. 4).

4. Discussion

Magmatic Ni-Cu-PGE, Ni-Cu-(PGE), and PGE-(Cu)-(Ni) deposits form through the concentration of large (former two) and small (latter) amounts of immiscible sulfide melts. Small amounts of sulfide melt can exsolve from magmas during crystallization, which is the likely source of the small amounts of sulfides in PGE-(Cu)-(Ni) deposits, but the larger amounts required to form Ni-Cu-PGE and Ni-Cu-(PGE) deposits must be incorporated by thermomechanical erosion of S-bearing crustal rocks country rocks during ascent and/or emplacement (e.g., Lesher et al., 1984; Lesher and Groves, 1986; Lesher, 1989; Lesher and Keays, 2002; Naldrett, 2004, 2010; Arndt et al., 2005; Barnes and Lightfoot, 2005; Keays and Lightfoot, 2010; Ripley and Li, 2013). Silicate components are normally assimilated, but the solubility of sulfide is very low, so the sulfides are left as xenomelts (Lesher and Campbell, 1993; Lesher and Burnham, 2001). The assimilated silicate components may have a small effect on sulfide solubility and a small to intermediate effect on the metal budget, but is normally not a governing process in generating sulfide-rich Ni-Cu-PGE or Ni-Cu-(PGE) deposits.

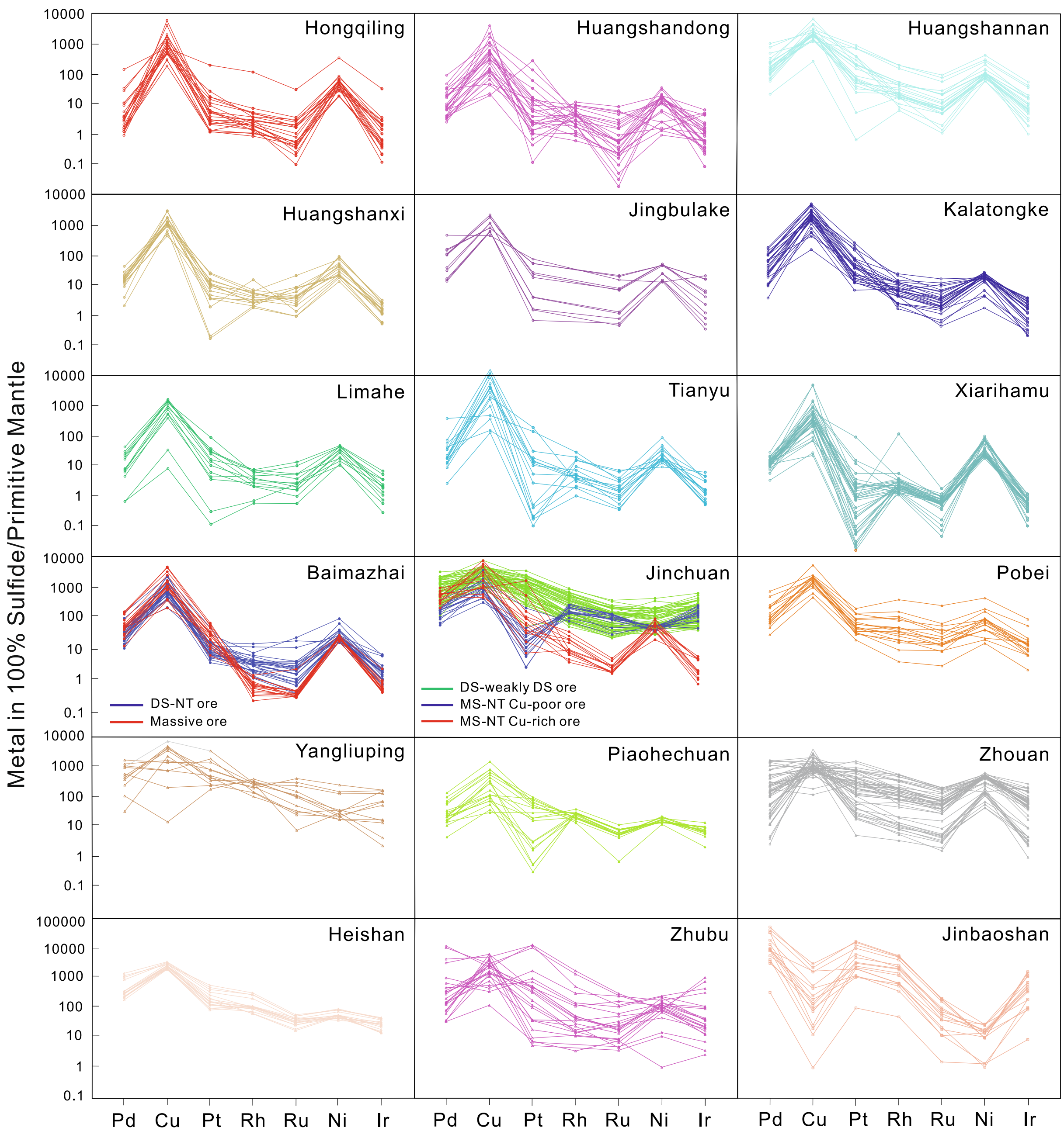

Fig. 10. Primitive mantle-normalized chalcophile element patterns for Chinese Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits. Concentrations recalculated to 100% sulfide using normalization values from Barnes and Maier (1999). Data sources as in Fig. 9.

A fundamental requirement of a system capable of thermo-mechanical erosion is a high-flux magmatic plumbing system (e.g., Lesher et al., 1984; Lesher, 1989; Barnes et al., 2016). Such systems are frequently, but not exclusively, fed by mantle plumes (e.g., Arndt et al., 2005; Barnes and Lightfoot, 2005) and localized near craton margins (e.g., Begg et al., 2010) in continental rift or rifted continental margin settings.

The compositions of the ores are controlled by 1) the metal abundances of the mantle source and the degree of partial melting (e.g., Naldrett and Barnes, 1986; Arndt et al., 2005; Maier et al., 2009), 2) the degree of fractional crystallization ± crustal contamination experienced by the magma during ascent (e.g., Lesher et al., 2001; Barnes et al., 2008), 3) the amount of magma that equilibrates with the sulfide (e.g., Campbell and Naldrett, 1979; Naldrett, 1981; Lesher and Burnham, 2001), 4) the degree of fractional crystallization of MSS (e.g., Hawley, 1962; Ebel and Naldrett, 1997; Barnes and Naldrett, 1987), and in some cases 5) interactions with magmatic-hydrothermal fluids (e.g., Hinchey and Hattori, 2005; Su and Lesher, 2012) or metamorphic-hydrothermal fluids (Lesher and Keays, 1984; 2002).

Below we discuss these processes as they apply to Ni-Cu-(PGE) and PGE-(Ni)-(Cu) deposits in China.

4.1. Temporal and tectonic settings

Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China have formed during over a relatively wide range of time in at least three fundamentally different tectono-magmatic settings:

1) 828–637 Ma Plume-Initiated Rifting and Breakup of Rodinia: The largest semi-continuous Ni-Cu ± PGE deposit in the world, Jinchuan, appears to have formed during the initial stages of breakup of the Chinese portion of the supercontinent Rodina, which

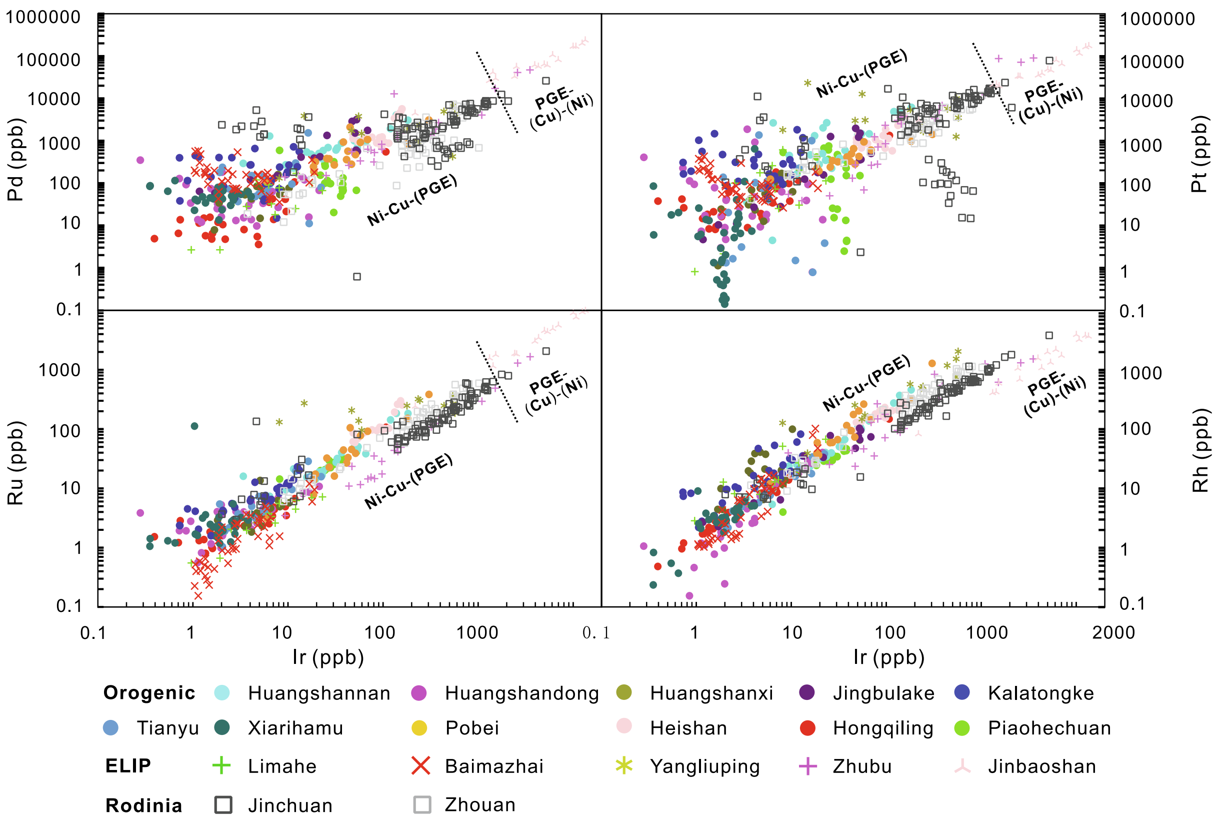

Fig. 11. Pd, Pt, Rh, and Ru in 100% sulfides vs Ir in 100% sulfides. Calculation method in text. Data sources as in Fig. 9.

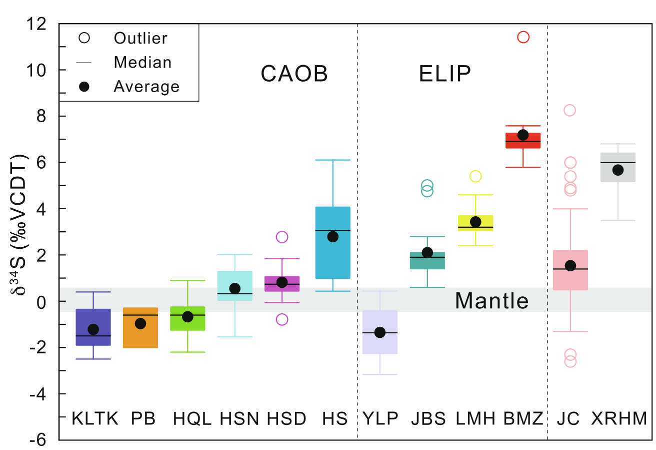

Fig. 12. Box and whisker plot of $\delta^{34}S$ for parts of the Chinese magmatic sulfide deposits. Coloured bars span range, black bar indicates median value, coloured box span 25th and 75th percentiles (second and third quartiles); circles are outliers (1.5*(Q3-Q1)) and far outliers (3*(Q3-Q1)). YLP Yangliuping; KLTK Kalatongke; PB Pobei; HQL Hongqiling; HSN Huangshannan; HSD Huangshandong; JC Jinchuan; JBS Jinbaoshan; HS Heishan; LMH Limahe; XRHM Xiarihamu; BMZ Baimazhai. Data sources: Ripley et al. (2005), Zhang (2005), Tao et al. (2007), Wang et al. (2010), Sun (2011), Dong et al. (2012), Jiang et al. (2012), Sun (2013), Xie et al. (2014), You (2014), Li et al. (2015), and Zhao et al. (2016a).

has been proposed to have been initiated by a superplume that produced mafic dikes and sills extending into South China and (then adjacent) south-central Australia (e.g., Li et al., 1999, 2008; Li et al., 2005; Xu et al., 2005; Xu et al., 2016; Pirajno, 2012; Wang et al., 2013a). The much smaller Zhouan deposit formed during the later stages of breakup, perhaps during a phase of lower magma flux. It is possible – if not likely – that much more mineralization formed during this event and remains to be found, but the breakup and dispersion of Rodinia may have destroyed much of it through erosion and subduction.

2) 431, 406, 357, and 287–269 Ma Arc-Related Magmatism in the Central Asian and East Kunlun Orogenic Belts: Larger numbers of smaller deposits were generated over an equally wide range of time during the complex history of the CAOB. There is some debate whether the mafic magmatism resulted from a mantle plume (e.g., Tarim LIP: Qin et al., 2011; Yuan et al., 2012; Yang et al., 2013; Yu et al., 2017) or arc magmatism (e.g., Li et al., 2012, 2015; Xie et al., 2014; Mao et al., 2016), but the latter seems to be gaining favour and is more consistent with the small sizes of the deposits and the geochemical and isotopic signatures of the host rocks.
3) 263–259 Ma Plume-Related Magmatism in the ELIP: This event produced several deposits over a more restricted time interval. The relatively small ELIP has been proposed to be related to the much larger Siberian LIP, which generated the supergiant Noril'sk-Talnakh camp (see review by Naldrett, 2004). The reasons for the differences in endowment may be magma flux (higher at Noril-sk-Talnakh, lower at Emeishan) and/or accessibility to crustal S (S-rich evaporites at Noril'sk-Talnakh vs S-poor Yangtze upper crust at Emeishan).

All three tectonic settings contain Ni-Cu-(PGE) deposits and may well all contain PGE-(Cu)-(Ni) deposits, although only the large Jinbaoshan PGE-(Cu)-(Ni) deposit has been discovered thus far. The range of ages and multiple tectonic settings highlight the fact that Ni-Cu ± PGE deposits formed throughout geological time in more than one tectonic setting (e.g., Arndt et al., 2005; Barnes and Lightfoot, 2005; Naldrett, 2004, 2011; Ripley and Li, 2013; Lesher, 2019). The deposits in China formed during continent assembly (CAOB and EKOB), during the initial (Jinchuan) and (Zhouan) final stages of continental breakup, and in an environment where the continent was not completely disassembled (ELIP deposits). A fertile magma in a high-flux conduit (lava channel or magma conduit) encountering a S source does not care about the tectonic setting or when it forms in a tectonic cycle, as long as those key ingredients are satisfied.

83% of the deposits in Table 1 occur within ~120 km of the margins of the North China and Yangtze cratons, and 61% occur within ~90 km of the margins, which is consistent with the roots of the cratons “steering” mantle plumes toward their margins (e.g., Kerrich et al., 2005; Sleep, 2005; Begg et al., 2010). There are other models that explain this association (e.g., edge-driven mantle convection: King and Anderson, 1998), but breakup of the entire supercontinent of Rodinia seems difficult to explain without a plume. There are also many deposits in China that occur in orogenic belts, so it is clear that plumes and rifting are not necessary to form Ni-Cu-(PGE) or Ni-Cu-PGE deposits, however, most are smaller and lower grade than the deposits in continental rifts (e.g., Duluth, Kambalda-Mt Keith-Perseverance, Noril’sk-Talnakh) and along rifted continental margins (e.g., Pechenga, Raglan, Thompson) probably because of lower magma fluxes and possibly lower amounts of crustal S.

4.2. Mantle source compositions

Geochemical and Nd-Os isotopic data indicate that the mafic–ultramafic magmas that formed most world-class magmatic Ni-Cu-PGE (e.g., Kambalda-Mt Keith-Perseverance, Noril’sk-Talnakh, Raglan, Thompson) and PGE-(Cu)-(Ni) (e.g., Bushveld, Great Dyke, Stillwater) deposits were derived by partial melting of HILE-depleted peridotitic (i.e., normal convecting asthenospheric) mantle (e.g., Lesher and Stone, 1996; Lesher et al., 2001; Arndt et al., 2005). The composition of peridotitic mantle varies with pressure, but generally consists olivine with lesser orthopyroxene and clinopyroxene, an aluminous phase (spinel, garnet, or plagioclase depending on pressure), and small amounts of Fe-Ni-Cu-(PGE) sulfide (e.g., Naldrett and Barnes, 1986; Arndt et al., 2005). The mafic–ultramafic host magmas that host magmatic Ni-Cu-PGE deposits typically formed at moderate to high degrees of melting, leaving olivine ± minor Opx as the only residual phase(s) (e.g., Naldrett and Barnes, 1986). Because Ni-Co-IPGE partition strongly into olivine and Cu-PPGE-Au partition strongly into the melt, the Ni-Co-IPGE and Cu-PPGE-Au contents of melts derived from peridotitic mantle increase and decrease with degree of partial melting, respectively (e.g., Naldrett and Barnes, 1986; Barnes and Naldrett, 1987).

However, if the olivine in peridotitic mantle is consumed by reaction with melts derived from recycled (e.g., subducted) ocean crust, the mantle will be pyroxenitic (Sobolev et al., 2005, 2007, 2009). Melts derived from pyroxenitic mantle will be characterized by enrichments in Ni-Co (because they will not be held back in olivine during partial melting) and Cu (which appears to be added metasomatically: Tonnelier, 2010) relative to PGE (which are depleted in MORB: Crocket, 2002). This suggests that the Ni-Co-Cu enriched, PGE-depleted patterns of many if not all of the Ni-Cu-(PGE) deposits in China (Fig. 13) are attributable to melting of pyroxenitic mantle (e.g., Tonnelier, 2010; Xu et al., 2013; Xie et al., 2013; Zhao et al., 2016b; Song et al., 2016).

Pyroxenitic and eclogitic mantle are thought to contain significantly more radiogenic Os than peridotitic mantle (Pearson and Nowell, 2004; Luguet et al., 2008) and if the mantle incorporated pyroxenetic melts the magma will be characterized by high γOs values (Lassiter et al., 2000; Huang et al., 2017). All of the Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China have significantly higher γOs(t) than deposits derived from peridotitic mantle (e.g., Raglan, Alexo, Kambalda, Thompson: Fig. 8) and plot on a mixing trend between magmas derived from pyroxenitic mantle and continental crust, suggesting these deposits were derived from a pyroxenitic source.

To further evaluate mantle source compositions we have plotted the Fo and Ni contents of olivine in several Chinese magmatic sulfide deposits and some typical komatiite Ni-Cu-(PGE) deposits in Fig. 14**A**, and have modeled the compositions of olivines crystallized from a magma derived from a peridotitic source containing 18% MgO, 10.6% FeO_t, and 526 ppm Ni (Straub et al., 2008) and from a magma derived from a pyroxenitic source containing 10% MgO, 8.6% FeO_t, and 540 ppm Ni (Straub et al., 2008). The olivine/magma partition coefficient for Ni is assumed to be 7 for the magma derived from a peridotitic source (Li et al., 2003a) and 12 for the magma derived from a pyroxenitic (Straub et al., 2008). We have also modeled removal of sulfide, which will lower Ni at a given Fo content (e.g., Duke and Naldrett, 1978; Lesher and Stone, 1996), regardless of the source. It is possible to enrich the Ni content of a magma by assimilating Ni-Cu-rich sulfides (not shown), but this would normally also result in proportionately high PGE contents.

Olivine in komatiite-associated Ni-Cu-PGE deposits and barren equivalents fall on or near the model for a peridotitic source in the Fo-Ni diagram (Fig. 14**B**). However, the olivines in many Chinese magmatic sulfide deposits (particularly Heishan, Huangshannan, and Zhouan) have relatively high Ni contents at intermediate-low Fo contents, well above the 10% MgO pyroxenitic magma fractional crystallization curve (Fig. 14**C**). Olivines at Huangshanxi, Tianyu, and Xiarihamu have lower Ni contents consistent with removal of Ni in sulfide, as noted by Tang et al. (2009a), Zhang et al. (2011), and Li et al. (2015), complicating a determination of the Ni content of the source.

There are other factors other than source composition than can influence the Ni contents in olivine, including upgrading via olivine-sulfide Fe-Ni exchange (see e.g., Brenan, 2003), variations in Si content (e.g., Wang and Gaetani, 2008), differences in melting and crystallization temperatures (e.g., Hart and Davis, 1978; Kinzler et al., 1990; Matzen et al., 2013), and differences in crustal magmatic processes (e.g., Herzberg et al., 2014, 2016). Lynn et al. (2017) examined the effects of crustal process (fractional crystallization, magma mixing, diffusive re-equilibration) on the Ni content in olivine from Hawaiian basalts and suggested that diffusive re-equilibration, crystal size, and sectioning effects can strongly affect the characterization of mantle source lithologies. However, taken together with the other geochemical and isotopic evidence presented above and discussed below, the olivine compositions are consistent with many of the magmas having been derived from pyroxenitic sources. This is consistent with the relatively young age of Chinese deposits in environments where there were many opportunities to metasomatize the underlying mantle (Fig. 15).

4.3. Fractional crystallization and crustal contamination

Determining the degree of fractionation crystallization and establishing primary magma compositions is difficult in mafic–ultramafic systems dominated by cumulate rocks, however, most cumulus minerals contain low to very low abundances of HILE, and Sr-Nd isotopes do not fractionate significantly during fractional crystallization, so they can be used to assess the degree of contamination. Because assimilating other materials consumes heat and because most magmas are not superheated (komatiites being a probable exception: Lesher and Groves, 1986), assimilation normally involves fractional crystallization. The maximum assimilation:fractionation ratio varies from ~0.3 in basalts (DePaolo, 1981) to ~0.5 in high-Mg komatiites (Lesher and Arndt, 1995).

Because magmas derived by partial melting of depleted peridotite (e.g., N-MORB), slightly-enriched peridotite (e.g., E-MORB), and moderately-enriched peridotite (e.g., OIB) are progressively enriched in Th relative to Nb and Yb, and because these elements are least-mobile during metamorphism, the Nb/Yb vs Th/Yb diagram (Fig. 16) provides a powerful way to evaluate 1) the sources of mantle-derived magmas (which will be reflected in similar degrees of Nb-Th enrichment), 2) the degree of subduction-related enrichment (which will be reflected in systematic enrichment in Th relative to Nb in all rocks from a given area), and 3) the degree of assimilation of continental crustal rocks (which will be reflected in variable degrees of enrichment in Th relative to Nb).

Most samples in the orogenic belts (CAOB and EKOB) have consistently high Th/Yb ratios and low Nb/Yb ratios (Fig. 16**A**). The

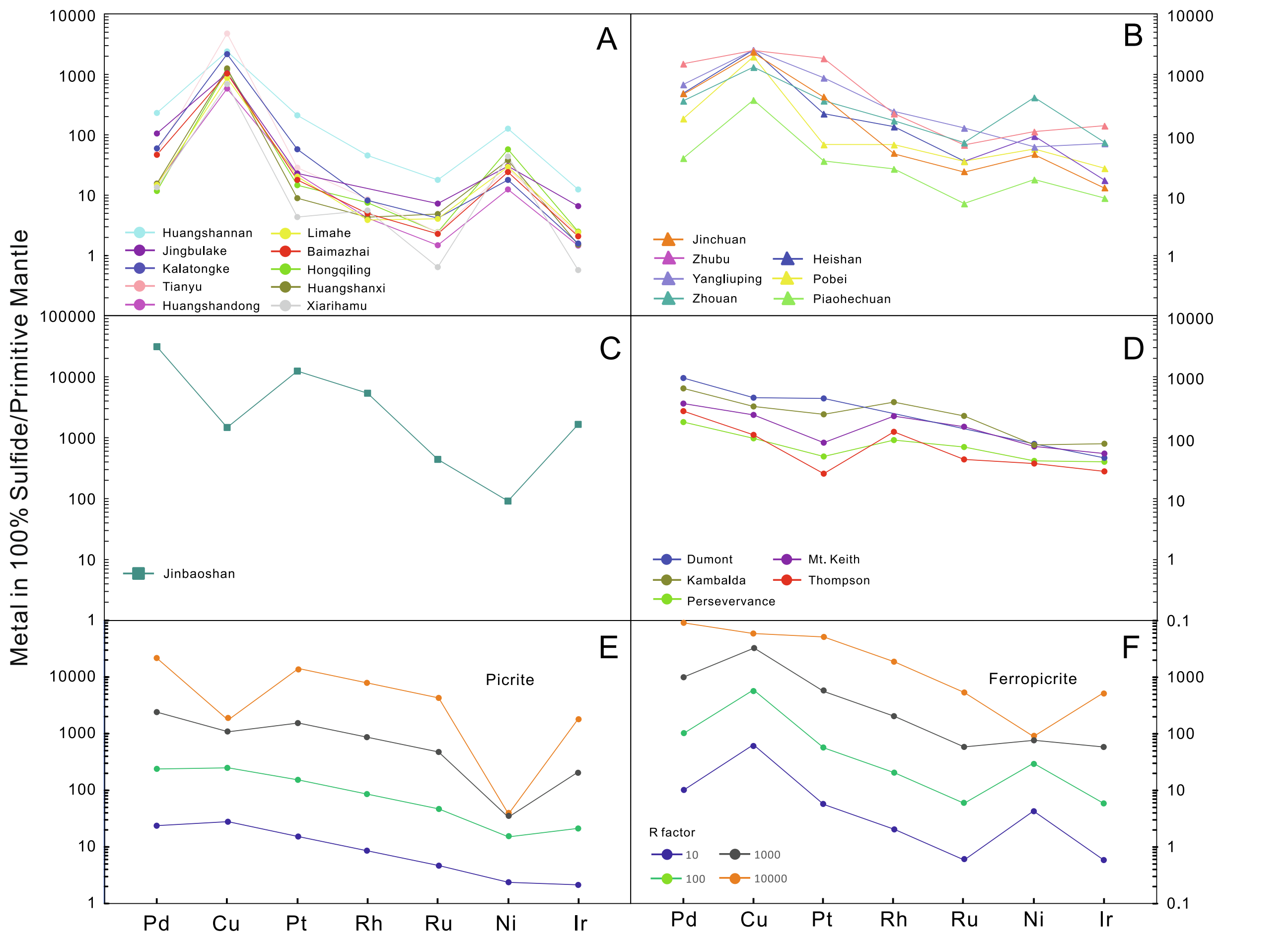

Fig. 13. Primitive mantle-normalized chalcophile element patterns with elements in order of increasing compatibility during mantle melting (Lesher and Keays, 2002). A) and B) Chinese Ni-Cu-(PGE) deposits; C) Chinese PGE deposit; D) komatiite-associated Ni-Cu-(PGE) deposits, E) modeled patterns for ores derived from a typical picrite at various R factors, and F) modeled patterns for ores derived from a typical ferropicrite at various R factors. Note that variations in R factor can increase PGE relative to Ni-Cu-Co, but cannot decrease PGE relative to Ni-Cu-Co. Data sources: Chinese deposits from references in Fig. 9; komatiite-associated deposits data from Lesher and Keays (2002); picrite composition and $D^{Sul/Sil}$ values from Barnes and Lightfoot (2005); ferropicrite composition and $D^{Sul/Sil}$ values from Barnes et al. (2001).

amount of AFC required to generate such high Th/Yb ratios is too great for most of the samples (they would not still be basaltic; Fig. 16**B**) and it would need to be relatively constant to form an array parallel to the mantle array. Together, they suggest derivation from depleted-enriched mantle that has been enriched in Th by subduction-related fluids/melts with minor amounts of crustal contamination. Models by Pearce (2008) suggest that 3–5% subduction component is required (Fig. 16**C**). The host rocks at Heishan, Huangshandong, Huangshanxi, Huangshannan, Pobei, Jingbulake, Hongqiling, and Xiarihamu appear to have crystallized from magmas derived from subduction-modified depleted (N-MORB-like) mantle, whereas the host rocks at Jinchuan, Zhouan, Kalatongke, Piaohechuan, and Baimazhai appear to have crystallized from magmas derived from subduction-modified enriched (E-MORB-like) mantle and to have incorporated variable amounts of continental crust. The host rocks at Jinchuan have extremely high Nb/Yb and very high Th/Yb ratios and anomalously low εNd(t) and high $^{87}Rb/^{86}Sr$, so have likely also been contaminated by rocks more evolved than upper continental crustal rocks, such as the adjacent marbles (Lehmann et al., 2007; Tonnelier, 2010; Song et al., 2012). The host magmas at Jinbaoshan, Limahe, Yangliuping, and Zhubu in the ELIP plot subparallel to the mantle array with slightly high Th/Yb and very high Nb/Yb. On this diagram they appear to be derived from a more enriched magma, possibly formed by interaction between an enriched plume source and a small amount of subducted crust, with minor crustal contamination.

Sr and Nd isotopic data provide additional constraints. The host rocks in the CAOB have positive $\varepsilon Nd_{(t)}$ values, suggesting that they formed from magmas derived from depleted mantle sources and experienced only minor crustal contamination. ELIP flood basalts have lower $\varepsilon_{Nd}(t)$ and higher $^{87}Sr/^{86}Sr_{(i)}$ values, suggesting that they were derived from a more enriched source than those in the CAOB. The host rocks in the ELIP have lower $\varepsilon_{Nd}(t)$ and trend to higher $^{87}Sr/^{86}Sr_{(i)}$, and have been interpreted to have experienced greater degrees of crustal contamination with Yangtze upper/middle crust (Zhou et al., 2008; Wang et al., 2012). Sr-Nd isotopes indicates that the degree crustal contamination degree for deposits in the ELIP is Baimazhai > Limahe > Jinbaoshan, consistent with differences in S isotopes (Fig. 7, Fig. 12). Thus, variations in both degree of source enrichment and crustal contamination have produced the differences in trace element and S-Nd-Sr isotopic compositions of deposits in the ELIP and CAOB, respectively. Host rocks related to the break-up of Rodinia (e.g., Jinchuan and Zhouan) exhibit a different mantle enrichment signature (toward EMI) and crustal contamination signature (toward lower continental crust). This has been suggested to represent involvement subcontinental lithosphere mantle (Li and Ripley, 2011; Wang et al., 2013b), but a significant contribution from SCLM is not supported by Os isotopic data (Fig. 8).

4.4. Parental magma compositions

Determining the major element compositions of parental magmas is

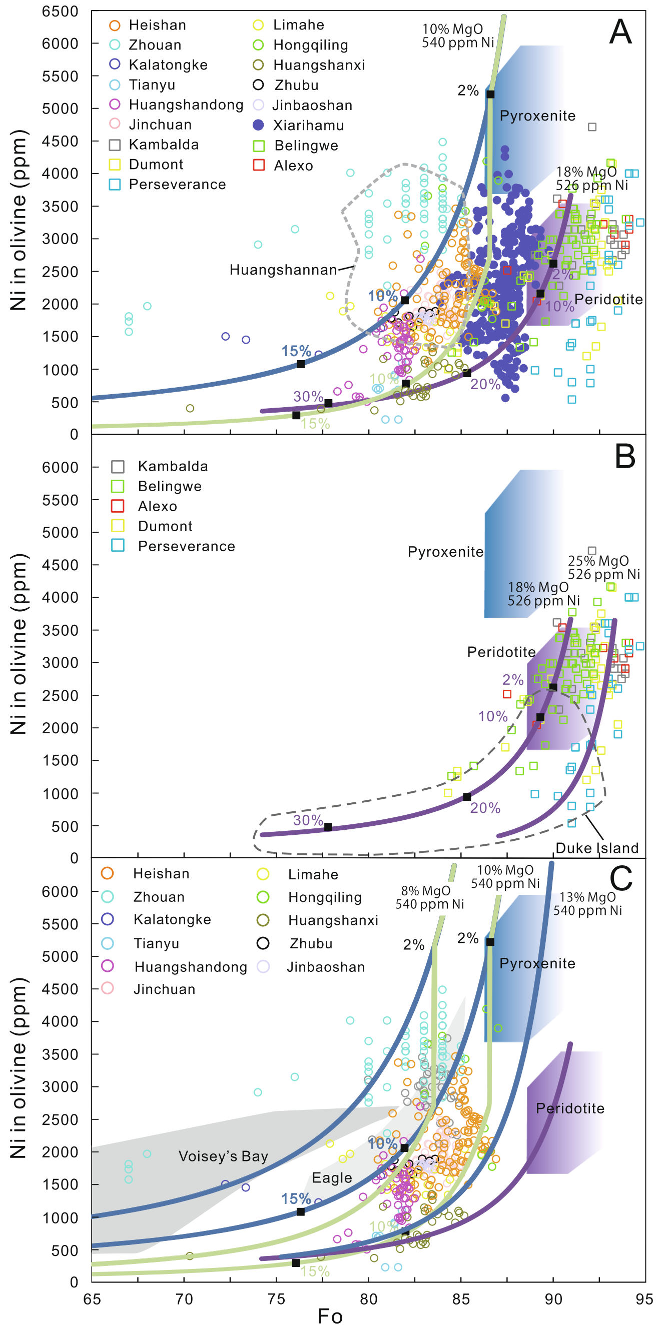

Fig. 14. Ni vs forsterite contents (Fo) of olivine (Ol) in Chinese magmatic sulfide deposits (A, C) and several komatiite-associated Ni-Cu-(PGE) deposits (A-B) for comparison. Blue curve represents fractional crystallization of Ol from a magma derived from a pyroxenitic source; green curve represents 2% crystallization of Ol from the same parental magma, followed by crystallization of Ol and segregation of sulfide (olivine:sulfide = 50:1); purple curve represents fractional crystallization of Ol from a magma derived from a peridotite source. Pyroxenite melts and peridotite melt compositions are from Straub et al. (2008, 2011). Data sources: Chinese deposits (A, C) from Li et al. (2004, 2012, 2015), Tao et al. (2007, 2008), Tang et al. (2009a), Zhang et al. (2011), Deng et al. (2012), Lu et al. (2012), Wang and Wang (2012), Xie et al. (2013), Zhao et al. (2016b); Duke island deposit data from Thakurta et al. (2008); Eagle deposit from Ding (2010); Voisey's Bay from Li and Naldrett (1999); komatiite-associated deposits from Lesher and Stone (1996).

also difficult in mafic–ultramafic systems dominated by cumulate rocks and many of the samples in the literature contain sulfides but have not been analyzed for S, precluding corrections for Fe in sulfide. Filtering for samples containing < 0.5 wt% Ni + Cu helps, but does not completely eliminate this problem. Even at Jinchuan, which has been the topic of several detailed studies and for which S data are available, it is not clear if the parental magma was a high-Mg basalt with 11.5% MgO and 11.2% FeO (Chai and Naldrett, 1992) or 12.3% MgO and 12.4% FeO (Li and Ripley, 2011), or a ferropicrite with 15.4% MgO and 13.7% FeO (Tonnelier, 2010). Part of the problem is that many studies have used regressions or averages, but that assumes all of the variations are sampling or analytical errors.

A detailed analysis at each location is beyond the scope of this contribution, but the key to constraining parental magma compositions in ultramafic rocks is to recognize that most samples will be derived from fractionated liquids and to identify the samples that must have accumulated the most magnesian olivine which must have crystallized from the most magnesian liquid. In this case, all areas contain at least some samples that define or lie on olivine-chromite-only control lines (Figs. 4, 5, and 17), indicating that most of the host rocks were derived from chromite-saturated picritic to high-Mg basaltic magmas. Only Xiarihamu, which contains high-Mg low-Cr (Ol-only) cumulates (Fig. 5) and highly magnesian olivine (up to Fo_{90}: Li et al., 2015), appears to have been derived from a chromite-undersaturated magma (see discussion by Lesher and Stone 1996). This requires higher degrees of partial melting and/or a source with lower Cr contents than the other deposits.

4.5. Host unit olivine ± opx enrichment and degree of differentiation

The host units range from poorly differentiated adcumulate-mesocumulate units with weighted bulk compositions that are strongly enriched in olivine ± Opx relative to their parental liquids to strongly differentiated units with thin lower cumulate zones and thick upper gabbroic-dioritic zones (Fig. 17, Table 2).

The bodies that are poorly differentiated and enriched in olivine (e.g., Jinchuan) have been interpreted by some authors (e.g., Tang 1993; de Waal et al. 2004) to have formed from olivine-sulfide rich “mushes”. However, olivine and sulfide are very dense and olivine-sulfide loaded magmas would not be capable of rising through the crust (see Lesher, 2017). It is more likely that these units – like those in many other Ni-Cu ± PGE systems – represent magma conduits in which olivine ± Opx accumulated during flow (Lesher et al., 1984; Lesher, 1989; Barnes, 2006; Arndt et al., 2008). Such systems are typically more prospective for Ni-Cu ± PGE mineralization than unenriched, differentiated units because they represent systems with higher magma fluxes that are more capable of thermomechanically eroding country rocks “upstream” (to incorporate crustal S) and maintaining high magma:sulfide ratios (to upgrade metal tenors) (e.g., Lesher et al., 1984; Lesher and Keays, 2002; Arndt et al., 2008). The bodies that are less enriched in olivine ± Opx and more strongly differentiated may have also had high magma fluxes at an early stage and “ponded” and differentiated after ore formation but before much olivine ± Opx accumulated, but there is clearly a correlation between the degree of olivine ± Opx accumulation and total Ni resource in Chinese and other Ni-Cu ± PGE deposits.

4.6. Host unit geometry

Although the focus of this paper is on geochemistry, the host units of Ni-Cu ± PGE deposits in China exhibit a wide range of geometries, reflecting different modes of emplacement and therefore ore localization.

Jinchuan has been interpreted as a subvertical dike (Tang, 1993), a rotated sill (Lehmann et al., 2007; Tonnelier, 2010), and a subvertical elongate funnel (Lightfoot and Evans-Lamswood, 2015). We find the arguments for structural rotation by Lehmann et al. (2007) and the evidence for asymmetric differentiation (requiring the same sense of rotation) by Tonnelier (2010) compelling, so we have classified Jinchuan as a channelized sill. However, if it has not been rotated,

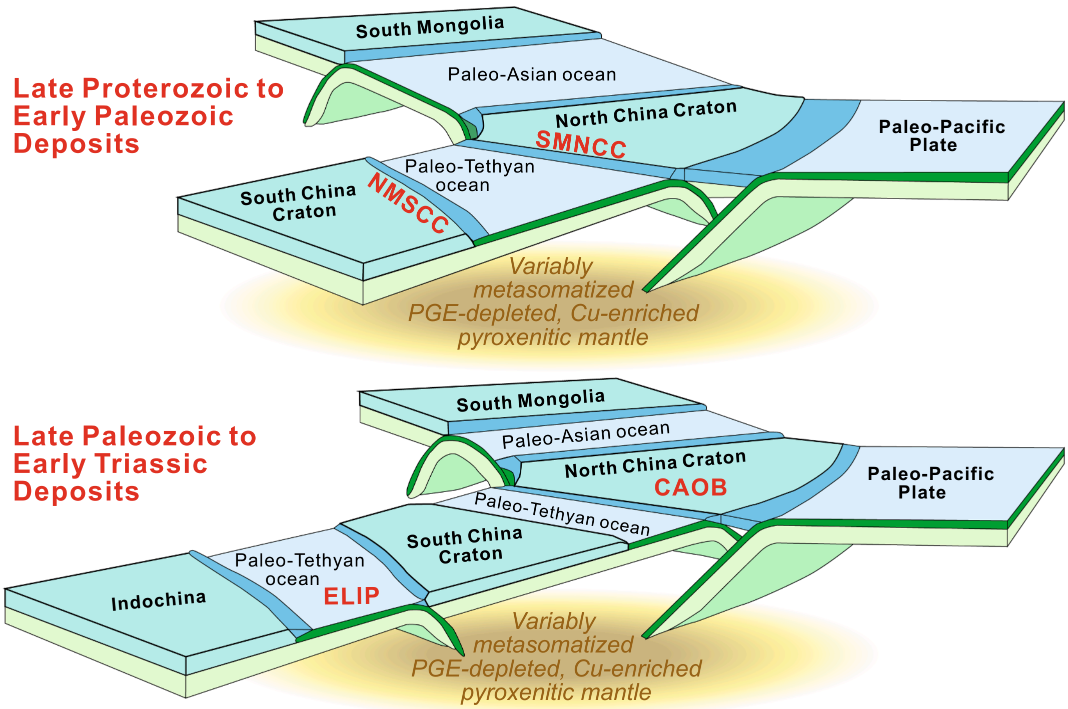

Fig. 15. Three-dimensional model showing tectonic settings (modified after Windley et al., 2010) and derivation of the host magmas for Chinese Ni-Cu-(PGE) deposits from subduction-modified pyroxenitic mantle sources.

Jinchuan is more likely a blade-shaped dike than a subvertical funnel (see discussion by Lesher, 2019).

Branquet et al. (2012) and Lightfoot and Evans-Lamswood (2015) classified several more of the bodies as subvertical funnels with diamond-shaped surface sections, but Hongqiling, Huanshannan, Huanshandong, Huanshanxi, Kalatongke, and Limahe are highly elongate subparallel to the surface, appear to close off at both ends, and are asymmetrically-differentiated. Although the former two aspects can be attributed to deformation, the latter cannot. For this reason, we have interpreted most of them as variably eroded subhorizontal chonoliths (Kalatongke #2) or blade-shaped dikes (remainder) (see also Barnes and Mungall, 2018). Only Jingbulake is asymmetrically zoned and may resemble a subvertical Alaskan-Uralian type intrusion.

The differences in the interpretations have important consequences for the mode of sulfide transport (subhorizontal rather than subvertical) and the sources of S (same stratigraphic level rather than deeper). More work is required to better distinguish between the different interpretations, but reinterpreting many of these bodies as subhorizontal channelized sills, blade-shaped dikes, and chonoliths eliminates all of the many problems inherent in transporting large abundances of dense sulfides vertically and suspending them in the throats of subvertical funnels while the host rocks crystallize (see Lesher, 2017; *in press*).

4.7. S Sources

Most magmas can dissolve and therefore exsolve only very small amounts of S (0.1–0.3%), so although very small amounts of sulfide (<1–2%) can exsolve during crystallization, larger amounts require the addition of external S (e.g., Lesher and Groves, 1986; Lesher, 1989; Lesher and Keays, 2002; Lightfoot and Keays, 2005; Keays and Lightfoot, 2010; Ripley and Li, 2013). S, Os, and Pb isotopes can be used to evaluate the degree of crustal S addition, although they are all susceptible to being overprinted by replenishment in dynamic magmatic systems (see Lesher and Stone, 1996; Lesher and Burnham, 2001).

Some of the deposits (Huangshannan, Huangshandong, Kalatongke, Pobei, Hongqiling) have $\delta^{34}S$ values close to the 0.1 ± 0.5‰ $\delta^{34}S$ range of MORB (asthenospheric peridotitic) mantle (see discussion by Lesher, 2017), but most exhibit variations well outside that range to more negative (e.g., Hongqiling, Kalatongke, Pobei, Yangliuping) or more positive values (e.g., Huangshandong, Huangshannan, Jinbaoshan, Jinchuan) or much more positive values (e.g., Heishan, Xiarihamu, Limahe, Baimazhai). Significantly, the ores in the CAOB and ELIP both range widely from slightly negative to moderately positive (CAOB) or strongly positive (ELIP).

The sulfides with near-mantle S isotopic compositions may have 1) exsolved from a unfractionated mantle-derived magma and been modified only slightly by interaction with fractionated crustal S, 2) incorporated unfractionated crustal S, or 3) incorporated fractionated S from a crustal source (discussed below) and exchanged S with the magma (e.g., Lesher and Stone, 1996; Lesher and Burnham, 2001; Ripley and Li, 2013), shifting their compositions varying degrees toward mantle. The low R factors deduced above for the sulfides at Huangshandong and Hongqiling suggest that the S isotopes in those deposits more likely approach that of their sources, whereas the higher R factors deduced above for Jinbaoshan, Yangliuping, and Jinchuan suggest that the S isotopes in those deposits more likely reflect exchange with the magma: in the case of Yangliuping an even lighter (more negative) source and in the case of Jinbaoshan an even heavier (more positive) source.

Given the trace element and Sr-Nd-Os isotopic evidence for a significant subduction component in the magmas that formed the host rocks at most of the deposits (e.g., Huangshandong, Huangshannan, Huangshanxi, Jingbulake, Tianyu, Xiarihamu) and for a significant crustal component in some of the deposits (e.g., Baimazhai, Jinchuan), there are at least three potential sources of non-mantle S: 1) isotopically heavy subduction-influenced arc mantle (e.g., Woodhead et al., 1987), 2) isotopically heavy continental crustal rocks, and/or 3) isotopically light continental crustal rocks. The latter is potentially the only option for Hongqiling, Kalatongke, Pobei, and Yangliuping, but the former two are difficult to distinguish. Trace element data suggest that the magmas in the CAOB and EKOB (e.g., Heishan, Hongqiling, Huangshandong,

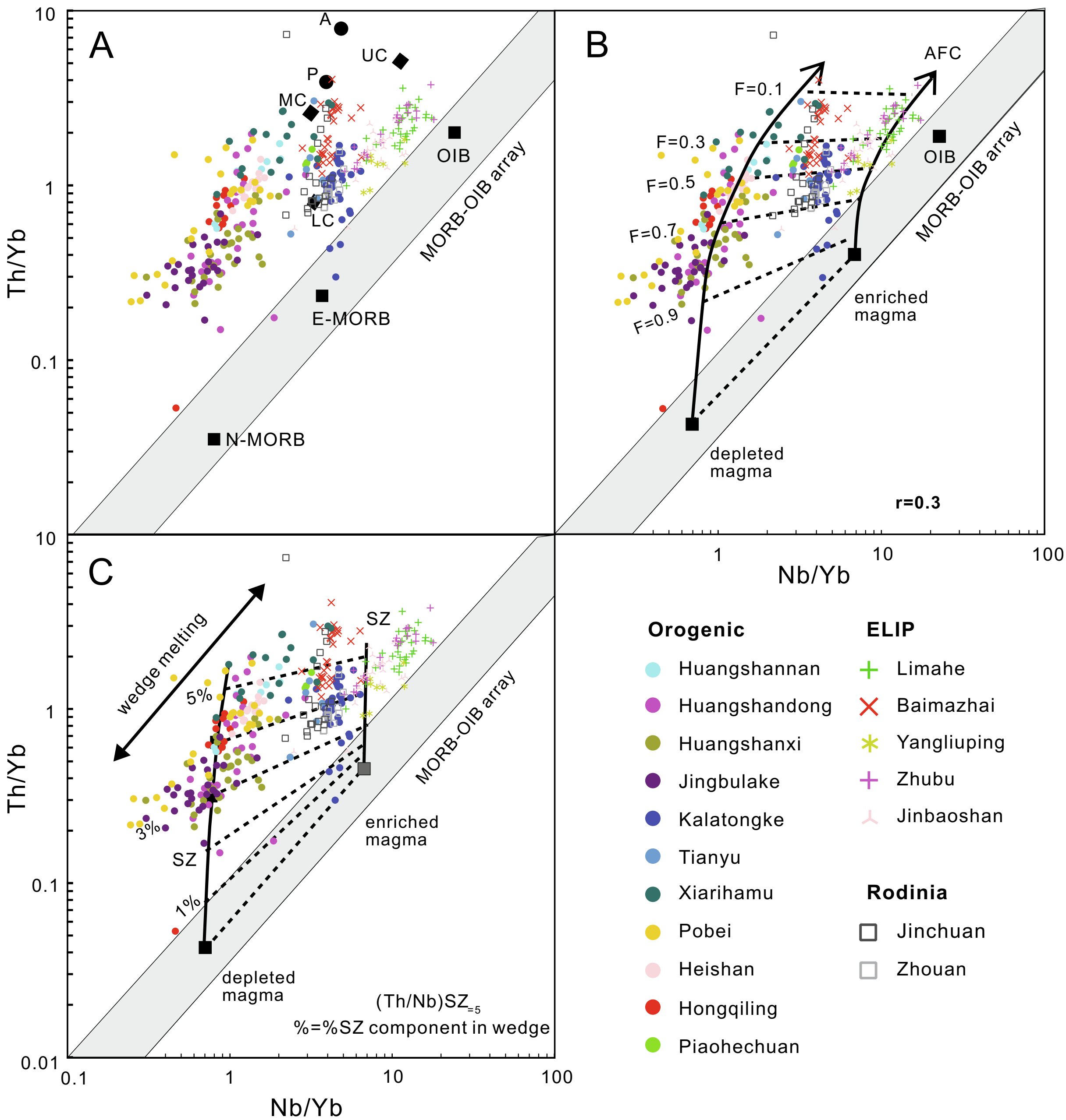

Fig. 16. Th/Yb vs Nb/Yb for A) Chinese deposits and major reservoirs, B) Chinese deposits and two model assimilation-fractionation crystallizations trajectories, and C) Chinese deposits and two models for subduction zone enrichment. SZ: subduction zone. Data sources in Fig. 4. Mantle array after Pearce (2008). N-MORB, E-MORB, OIB, and Primordial Mantle (PM) after Sun and McDonough (1989); average lower crust (LC), upper crust (UC), total continental crust (CC) and felsic Phanerozoic (P) and Archean (A) crust are from Rudnick and Fountain (1995).

Huangshannan, Kalatongke, Pobei) originated from subduction-modified arc mantle ($\delta^{34}S > 2‰$: Woodhead et al., 1987), so the ~ 0 or slightly negative S isotopes for some of these deposits (Hongqiling, Huangshandong, Huangshannan, Pobei) was likely caused by crustal contamination (Fig. 12) (see also Mao et al., 2018a).

The S and Os isotope geochemistry of the ores suggest that the Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China include S and metals derived from at least two sources. 1) Deposits in the ELIP are characterized by variable S and Os isotopes. These data can be interpreted in two ways: for the deposits that are sulfide-rich and appear to have formed at relatively low R factors, this suggests a crustal source, but for the deposits that are sulfide-poor and appear to have formed at high R factors, the source may be the mantle-derived magma or it may be a crustal source modified by reaction with the magma as discussed above. 2) Deposits in the CAOB and EKOB are characterized by lower PGE_{100} and higher $^{187}Os/^{188}Os$ ratios, and clearly incorporated crustal S. The sources of crustal S are not well understood, so this is an area where more research is required to determine whether the S came from deeper sources and was physically transported from depth, whether the S came from more-or-less the same stratigraphic levels and was transported only horizontally, or whether the S came from shallower sources and settled backwards (see discussion by Lesher, 2017; Lesher, 2019).

4.8. Ore localization

Most of the mineralization in Chinese Ni-Cu-(PGE) deposits occurs as low-grade disseminated sulfides, although some deposits (e.g., Baimazhai, Jinchuan, Xiarihamu) contain significant amounts of net-

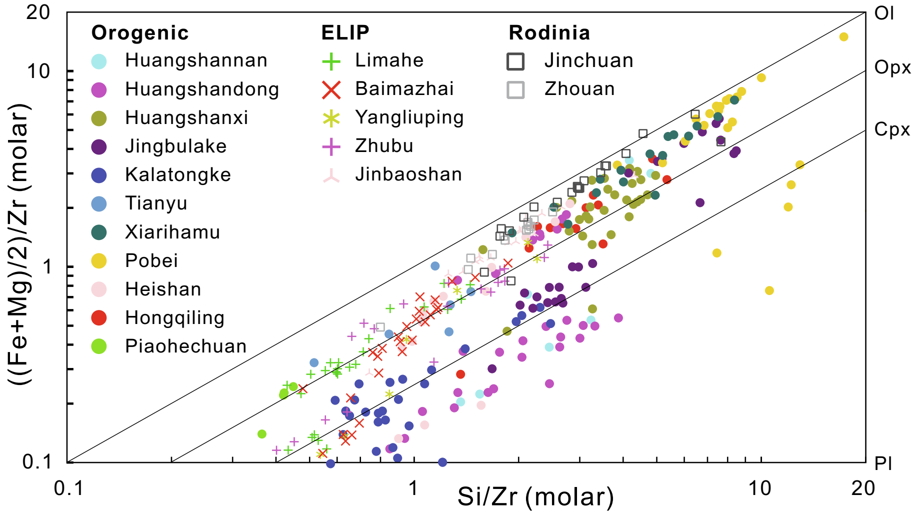

Fig. 17. Molar Si/Zr vs molar ((Fe + Mg)/2)/Zr. Data sources in Fig. 4.

textured sulfides and minor massive sulfides.

The different locations of the mineralized zones – basal/marginal vs. single internal vs. multiple internal – indicate that the mineralization formed during the early, intermediate, and intermediate-late stages of emplacement and crystallization of the host units (see Lesher and Keays, 2002) in different areas:

1) Early basal/marginal mineralization: lateral parts of Jinchuan, Heishan, Hongqiling, Limahe, central parts of parts of Piaohechuan, parts of Yangliuping, Zhubu
2) Intermediate internal mineralization: Baimazhai, Jingbulake, central parts of Jinchuan, Kalatongke, Pobei, lateral parts of Piaohechuan, parts of Yangliuping
3) Intermediate-late multiple internal mineralization: Huangshannan, Huangshandong, Huangshanxi, Jinbaoshan, Tianyu, Xiarihamu

Although some workers have suggested that massive ores have been emplaced later than disseminated-net textured mineralization, this likely reflects late mobility of massive sulfides, which remain molten at temperatures well below the crystallization temperatures of the silicate minerals (e.g., Tonnelier, 2010; Barnes et al. 2016) and therefore the networks of disseminated and net-textured ores.

The mode of sulfide emplacement is still being debated. Tang (1993), de Waal et al. (2004), Li et al. (2015), Song et al. (2016), Lightfoot and Evans-Lamswood (2015), and Wang et al. (2018) have suggested that the Ni-Cu-(PGE) deposits in China formed by mobilization of sulfide-rich magmas from deeper chambers. The motivation for such models appears to be 1) a paucity of S in local country rocks and/or 2) depletion of PGE relative to Ni-Co-Cu in the ores, which is interpreted to reflect prior removal of small amounts of sulfides. There are several problems with such models:

1) The S source need not be at depth, it may have been along strike in the plumbing system. Many of the deposits occur in sedimentary sequences that likely contained at least some sulfidic pelites or evaporites and many of the host units appear to have (or had) subhorizontal orientations (as discussed above).
2) Depletion of PGE relative to Ni-Co-Cu can also be attributed to the mantle source and parental magma being depleted in PGE and enriched in Ni-Co-Cu (as discussed above). However, a more serious problem with this model is that it requires unrealistically high R factors (up to 10^6 in the case of some elements) to model the observed ore compositions.
3) Sulfide melts are very dense ($\sim 4.2\ g\ cm^{-3}$) and very fluid ($\sim 0.1\ g\ cm\ sec^{-1}$) compared to silicate magmas ($\sim 2.7\ g\ cm^{-3}$ and $10–1000\ g\ cm\ sec^{-1}$), so any > 10–15% is impossible to transport in magmas ascending buoyantly through continental crust (ave $< 2.9\ g\ cm^{-3}$) (e.g., Lesher and Groves, 1986; Lesher, 2017).
4) It is possible for fine, dilute suspensions of sulfide droplets to be transported and collected on "filter beds" in overlying chambers (Lesher, 2017), but that does not explain why significant amounts of Ni-Cu ± PGE mineralization or PGE enrichment have not been reported in volcanic rocks other than in situations where there is evidence for it having formed at that stratigraphic level (e.g., Alexo, Kambalda, Raglan; see discussion in Lesher, 2017; *in press*).
5) It is possible for greater amounts of sulfide droplets to be transported in the presence of volatile bubbles (Mungall et al., 2015) or felsic xenoliths (Lesher, 2017), which can reduce the bulk density of the magma, but none of the deposits in China contain abundant vesicles like those observed at Noril'sk (e.g., Barnes et al., 2017; Lesher, 2017) or abundant xenoliths like those observed at Aguablanca (e.g., Tornos et al., 2001) or Voisey's Bay (e.g., Li and Naldrett, 2000).
6) It is feasible to transport sulfide-rich magmas via seismic pumping

Table 2
Variations in degree of olivine enrichment and degree of differentiation for Chinese Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits (adapted from Lesher et al., 1984; Lesher and Keays, 2002; Arndt et al., 2008).

	Poorly Differentiated	Weakly Differentiated	Strongly Differentiated
Strongly enriched in Ol ± Opx	Jinchuan Zhouan Xiarihamu Jinbaoshan Huangshannan Pobei		
Moderately enriched in Ol ± Opx		Zhubu Baimazhai Limahe Huangshanxi	
Unenriched in Ol ± Opx			Heishan Huangshandong Jingbulake Kalatongke

(Lesher, 2013a,b, 2017, in press) and Lightfoot and Evans-Lamswood (2015) have suggested that dense sulfide-rich magmas were pumped from depth through a subvertical conduit system undergoing alternating transtension and transpression, and localized in dilations and traps created by cross-linking structures in strike-slip fault zones. However, such a model does not explain the absence of any sulfides in areas where overlying volcanic equivalents are exposed (e.g., ELIP), some of which are consistent with being near-primary compositions that have not fractionated significantly *en route* to the surface and none of which are depleted in PGE (Lesher, 2019).

It is equally possible if not probable that the sulfides were generated at more-or-less the same stratigraphic level and were trapped at those levels (Lesher, 2017, *in press*).

The abundance of low-grade mineralization in many deposits can likely be attributed to a paucity of available S. Very high magma fluxes would also favour disseminated rather than net-textured or semi-massive mineralization (e.g., Lesher and Keays, 2002), but most of the low-grade systems do not appear to be olivine-rich high-flux systems.

4.9. R Factor

Although the abundances of Ni-Co-Cu in most mafic–ultramafic magmas are high enough that the small amounts of sulfide that typically exsolve from magmas may extract enough metals to generate reasonable metals tenors without severely depleting the magma, because the partition coefficients ($D_{Co}^{Sul/Sil} \sim 30$, $D_{Ni}^{Sul/Sil} =$ 100–500, $D_{Cu}^{Sul/Sil} =$ 500–1000, and $D_{PGE}^{Sul/Sil} = 10^5$-10^6) are so high (see reviews by Lesher and Stone, 1996; Crocket, 2002; Makovicky, 2002; Naldrett, 2004; Barnes and Lightfoot, 2005), there are limitations on how much Ni-Co-Cu can enter moderate-large amounts of sulfide and limitations on how much PGE can enter even very small amounts of sulfide. The mass balance varies, depending on whether sulfide is exsolved from the magma (Campbell and Naldrett, 1979), incorporated from a barren external source (Naldrett, 1981), or incorporated from an external source containing metals (Lesher and Burnham, 2001). The general form of the mass balance is (Lesher and Burnham, 2001):

$$Y_i^f = \frac{(X_i^o R + Y_i^o) D_i^{Sul/Sil}}{R + D_i^{Sul/Sil}} \quad [1]$$

where X_i^o is the initial concentrations of metal *i* in the silicate liquid, Y_i^o is the initial concentration of metal *i* in the sulfide melt, *R* is the magma/sulfide mass ratio, $D_i^{Sul/Sil}$ is the sulfide/silicate partition coefficient, and Y_i^f is the final concentration of metal *i* in the sulfide melt.

The major unknown is the amount of metal in the magma (X_i^o), which is better constrained in areas where chilled margins or contemporaneous volcanic rocks are present (e.g., ELIP), but poorly constrained in areas containing only cumulate rocks (e.g., Jinchuan). We are less confident than some authors in our ability to estimate these values, particularly given the spectrum of mantle sources indicated by the lithogeochemical, Ni in olivine, and Sr-Nd-Os isotopic data. For example, ores depleted in PGE relative to Ni-Cu-Co have been interpreted by some authors to have formed by prior extraction of small amounts of sulfide from a magma derived from a peridotitic source (Song et al., 2008; Gao et al., 2012, 2013; Zhao et al., 2015) and by other authors to have formed by equilibration with a magma derived from a Ni-Cu-Co-enriched, PGE-depleted pyroxenitic source (Tonnelier, 2010; Xu et al., 2013; Xie et al., 2013; Zhao et al., 2016b; Song et al., 2016). As noted above, the Os isotopic data are fairly definitive and in the case of Jinchuan the former model predicts *R* factors that are much too high (up to 10^6 in the case of Ir) for a such a sulfide-rich deposit, and the latter model predicts *R* factors that are much more reasonable (~500) (Tonnelier, 2010).

We have modelled R factor variations in a peridotite-derived picritic magma and a pyroxenite-derived ferropicritic magma in Fig. 13**E-F**. It is clear that variations in R factor can increase PGE relative to Ni-Cu-Co, but cannot decrease PGE relative to Ni-Cu-Co. As discussed below, crystallization of MSS from sulfide melt will fractionate Ni-Co-IPGE from Cu-Au-PPGE, not all PGE from Ni-Cu-Co. Thus, high R factors can erase or reduce the pyroxenite signature and MSS fractionation can modify the pyroxenite signature, but only a magma depleted in PGE > Ni-Cu-Co can produce an ore that is depleted in PGE > Ni-Cu-Co.

Ir is virtually insoluble in magmatic-hydrothermal and metamorphic-hydrothermal fluids (Keays et al., 1982; Lesher and Keays, 1984, 2002; Su et al., 2011) and nearly all samples have tight positive correlations in Ir vs Rh and Ir vs Ru diagrams (Fig. 11). The Jinbaoshan PGE deposit has the highest abundance of PGE in sulfides, suggesting a very high *R* factor (up to 10^5: Lu et al., 2014). Pobei and Zhouan have moderate PGE content in sulfides, consistent with moderate to high R factors (up to 6000: Wang et al., 2013b; Yang et al., 2014). Other Ni-Cu deposits in China have been proposed to have formed at low-moderate R factors (30–1000) resulting in systematic PGE depletion relative to Co-Ni-Cu (Tao et al., 2008; Zhang et al., 2009; Yang et al., 2012; Wei et al., 2013; Sun et al., 2013). However, the calculated *R* factors vary depending on the abundances of PGE in the chosen parental magma. A magma derived from a pyroxenitic source, for example, will have higher Ni-Cu-Co and lower PGE (e.g., Lesher and Stone, 1996; Tonnelier, 2010), increasing – with all else equal – the apparent *R* factor than for a magma derived from a peridotitic source.

The decoupling of Os from Sr-Nd isotopes in Jingbulake deposit has been attributed to selective incorporation of Os-rich, Sr-Nd-poor sulfides relative to Os-poor, Sr-Nd-rich silicate components (Yang et al., 2012). This seems logical given the lower melting point of sulfides, but there is no evidence for selective incorporation of sulfides relative to silicates in other deposits where this has been studied in detail (e.g., Duluth, Kambalda, Raglan, Thompson, Voisey's Bay). Furthermore, massive sulfides at Limahe and Jingbulake have high $\gamma Os_{(t)}$ values and low PGE tenors, both consistent with low R factors, and disseminated and sulfide-poor samples have low $\gamma Os_{(t)}$ values and high PGE tenors, both consistent with high R factors. It is more likely that the decoupling of Os isotopes from Sr-Nd isotopes can be attributed to their relative abundances in sulfide melt (Os >>> Sr-Nd) and silicate magma (Sr > Nd >> Os) (Lesher and Burnham, 1999, 2001). As a result of variable *R* factors, these deposits show variable $\gamma Os_{(t)}$ over a limited range of εNd (Fig. 8).

4.10. Fractional crystallization of sulfide liquid

Experimental studies have shown that the first phase to crystallize from a sulfide liquid is Fe-Ni-IPGE rich, Cu-PPGE-Au-poor monosulfide solid solution (MSS), leaving a residual sulfide melt that is enriched in Cu-PPGE-Au (e.g., Barnes and Naldrett, 1987; Fleet and Pan, 1994; Ebel and Naldrett, 1997; Sinyakova and Kosyakov 2009). In the Ir_{100} vs PGE_{100} plots (Fig. 11), most samples have tight positive correlations between Ir, Ru, and Rh because these elements have similar partition coefficients between MSS and sulfide liquid. Samples from Jinbaoshan and Huangshandong have lower Ru/Ir ratios, which may indicate crystallization of chromite, as Ru is known to partition into chromite (e.g., Brenan et al., 2012; Pagé and Barnes, 2016). In the Ir_{100} vs Pd_{100} and Pt_{100} plots (Fig. 11), however, quite a few samples depart from the positive trend. Pt_{100} is decoupled from Pd_{100} and decreases at more-or-less constant Ir_{100} in many Xiarihamu samples and some Jinchuan and Piaohechuan samples, suggesting that Pt (which commonly occurs as Pt alloys rather than in PGMs) has segregated or been mobile (see Song et al., 2009; Chen et al., 2013). Pd and Ir are inversely correlated significantly only at Jinchuan and possibly Zhouan, consistent with accumulation of MSS (samples with higher Ir and lower Pd) or residual sulfide liquid (samples with lower Ir and higher Pd). For the most part, most other deposits have relatively consistent Cu-Ni-Co patterns and

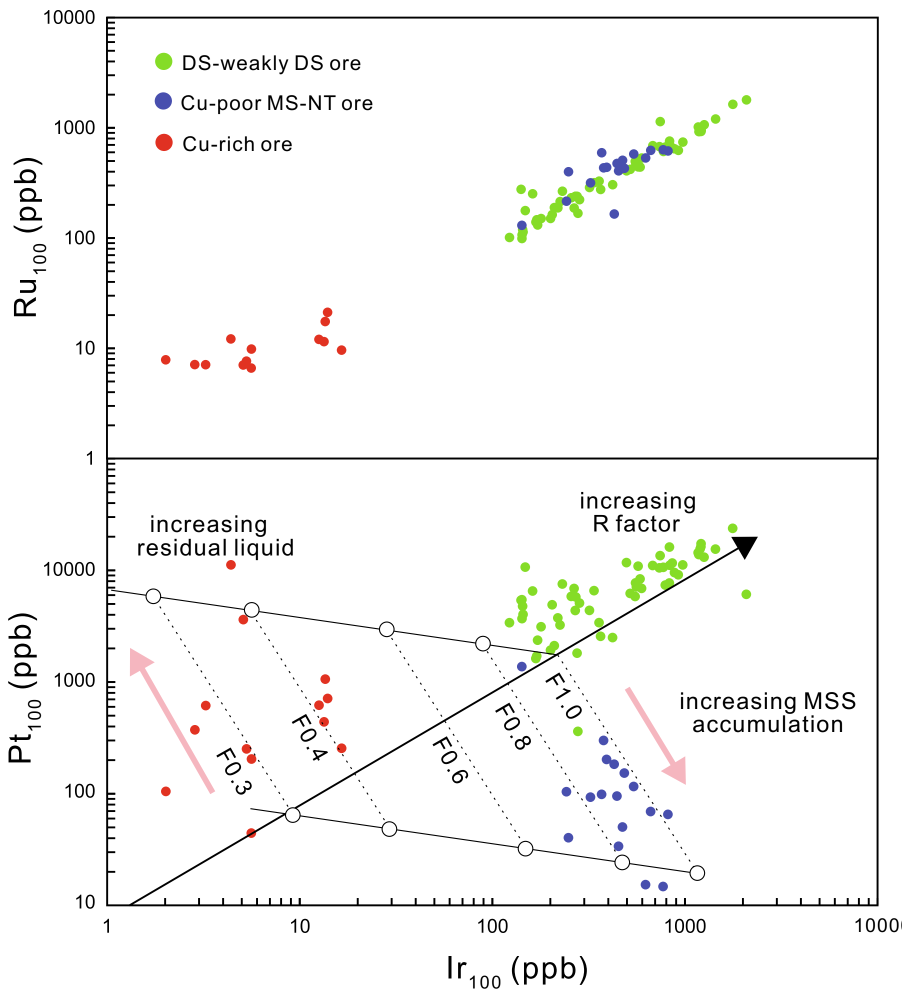

Fig. 18. Pt_{100} and Ru_{100} vs Ir_{100} vs for Jinchuan. Data from Chen et al. (2013). Modelling parameters: $D_{Ir}^{Sul/Sil} = 35000$, $D_{Pd}^{Sul/Sil} = 35000$; $D_{Ir}^{MSS/Liq} = 5$, $D_{Pd}^{MSS/Liq} = 0.01$.

relatively consistent PPGE-IPGE patterns. This suggests that the majority of the deposits have not fractionated significantly and/or that the sampling has homogenized any fractionation.

The finer details of MSS fractionation are well illustrated at Jinchuan. The positive correlation between Ir_{100} and Ru_{100}-Pt_{100} in disseminated ores (Fig. 18) indicates that the variations are controlled by magma:sulfide ratios (R factors), not MSS fractionation. However, some of the net-textured and massive ores are enriched in Fe-Ni-IPGE and depleted in Cu-PPGE-Au (i.e., enriched in MSS) and some are depleted in Fe-Ni-IPGE and enriched in Cu-PPGE-Au (i.e., enriched in residual sulfide melt). Thus, there are three different mantle-normalized PGE patterns for Jinchuan mineralized samples (Fig. 10) and only the disseminated ores reflect the composition of the original sulfide melts.

5. Conclusions

1) Nearly all Chinese Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits occur along or near the margins of cratons (e.g., North China craton, Yangtze Craton) or in orogenic belts (e.g., Central orogenic belt, East Kunlun orogenic belt) that likely represented craton margins, consistent with those locations representing areas where magmas were able to preferentially rise through the crust.
2) Olivine, chalcophile element, and Nd-Sr-Os isotopic compositions suggest that many host magmas formed by melting of metasomatized pyroxenitic mantle, likely produced by the interaction of recycled oceanic crust with variably-enriched peridotitic mantle, but the compositions of the parental magmas have been modified by minor to significant degrees of crustal contamination.
3) Although many deposits appear to have been derived from a pyroxenite source, the composition of the original peridotitic mantle, the degree of mantle metasomatism, and the degree of crustal contamination appear to have been different. Most deposits in the CAOB and EKOB appear to have been derived from magmas formed by partial melting a metasomatized but originally depleted mantle source with minor crustal contamination. Most deposits in the ELIP appear to have been derived from magmas formed by partial melting of more enriched sources than in the other metallogenic provinces. Those related to the breakup of Rodinia exhibit transitional geochemical characteristics.
4) The deposits appear to be hosted by subhorizontal blade-shaped dikes, chonoliths, and channelized sills, not subvertical funnels as proposed by other workers.
5) S isotopic data indicate a wide range of crustal S sources, modified by interaction with variable amounts of mantle-derived magmas.
6) The compositions of most Chinese Ni-Cu-(PGE) deposits have been modified to some degree – in some cases significant degrees (e.g., Jinchuan) – by MSS fractionation. Averaging samples without weighting them will not properly compensate for this process and even net-textured ores are susceptible to fractionation. Only disseminated ores are reliable indicators of original sulfide melt compositions, but care must be taken to ensure that low-grade samples have not experienced modifications in Ni contents accompanying

serpentinization of olivine.

7) The wide variations in tectonic-magmatic setting (breakup of Rodinia vs CAOB vs ELIP), host rock composition (dunite to diorite), geochemistry (HILE-enriched to HILE-depleted), and isotope geochemistry (high to low $^{87}Sr/^{86}Sr/\varepsilon Nd/\gamma Os$, positive to negative $\delta^{34}S$) indicate that these factors were not critical to the formation of magmatic Ni-Cu-(PGE) and PGE-(Cu)-(Ni) deposits in China, but that the key factors are channelized magma flow to thermo-mechanically erode country rocks upstream in the system, to access crustal S, and to achieve at least moderate R factors.

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